

A Spectral Criterion for Emergent Gravity

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ABSTRACT: This paper formulates a spectral criterion for emergent gravity from the energy-momentum tensor commutator. The Weinberg–Witten theorem excludes an elementary massless spin-2 particle in a Poincaré-invariant quantum field theory with a covariant conserved stress tensor. A massless spin-2 excitation, if it appears in the stress-tensor spectral function, must therefore be interpreted as a collective mode rather than as a fundamental field. Since the spectral function is defined by a commutator, additive static vacuum-energy terms do not contribute to the dynamical spin-2 spectral density. Coarse-graining in momentum space constructs, at each scale, a macroscopic rank-2 tensor field. The irreversibility of this operation promotes the scale itself to a dynamical order parameter. Its Langevin dynamics is governed by two renormalization-group fixed points, an ultraviolet repeller at the Planck scale and an infrared attractor at the Hubble scale. Entropy production and energy transfer define an effective temperature, and the fluctuation–dissipation theorem relates the corresponding noise and dissipation. The criterion advanced in this paper is whether the spin-2 spectral density can develop an isolated zero-momentum pole. Such a pole is absent in perturbation theory and represents a nonperturbative reorganization of spectral weight. If the pole is present, Weinberg’s low-energy theorem fixes the infrared effective action to Einstein–Hilbert form. Newton’s constant is then read from the pole residue, while the infrared fixed point supplies the natural curvature scale. The criterion is in principle testable. Lattice gauge theory or functional renormalization group methods can decide whether the spin-2 pole forms.

KEYWORDS: Emergent gravity; Spectral function; Spin-2 polarization; Langevin equation; Renormalization group fixed point; Fluctuation–dissipation theorem; Coarse-graining

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1 Introduction

The spectral representation of two-point correlation functions in the vacuum, the Källén-Lehmann representation [1], can be decomposed by spin into distinct polarization sectors. These sectors contain spin-2, spin-1, and spin-0 projection operators [2]. In the Standard Model, this spectral analysis framework is applied separately to quantum fields of different spin. The spectral function of gauge fields describes spin-1 propagation, and that of fermion fields describes spin-1/2 propagation. The energy-momentum tensor is a rank-2 symmetric tensor operator in quantum field theory. It is the conserved Noether current of spacetime translation symmetry. Its transversality condition eliminates the spin-1 sector, leaving only spin-2 and spin-0 contributions [3]. In the spectral function of the energy-momentum tensor, the spin-2 channel is the lowest nontrivial excitation channel that remains unsuppressed.

Several quantum field theory effects can be understood as modulations of the vacuum fluctuation spectral function under specific conditions. The Casimir effect originates from mode truncation of the electromagnetic vacuum fluctuation vector spectral function under boundary conditions [4, 5]. In electroweak theory, originally massless gauge and fermion fields acquire mass through coupling to vacuum condensates. The spectral function transforms from a massless continuum to a form with mass-threshold structure [6]. In QCD, the quark condensate signals spontaneous chiral symmetry breaking [7]. Viewed in terms of the spectral function, this corresponds to a nonperturbative reshaping of quark and gluon spectral functions [8, 9]. In Hawking radiation, the horizon imposes absorbing boundary

conditions on quantum field modes. From the perspective of an external observer, the Minkowski vacuum spectral function turns into a thermal spectral form [10, 11].

These examples involve modulations of the spectral function by external conditions. Even without special conditions such as boundaries or condensates, the vacuum spectral function itself carries structural information about the fundamental interactions. If we examine the fundamental interactions themselves from a similar perspective, their coupling effects should be regarded as encoded in the structure of the vacuum spectral function, rather than arising from specific external physical conditions that modulate it. This distinction is central to what follows. If the gravitational interaction is encoded in the spectral function, the question becomes what specific structure in $\rho^{(2)}$ signals its presence.

In quantum electrodynamics, the spectral function of the photon field two-point correlation function fully encodes the effects of the electromagnetic interaction. The free-field contribution appears as a $\delta(p^2)$ -type term, while interaction corrections manifest as a continuous p^2 -dependent component superimposed upon it. The strength of this component is proportional to the fine-structure constant α [12]. The photon spectral function captures the complete physical content of a fundamental massless spin-1 particle. Its form is determined by the structure of the electromagnetic coupling, and the behavior of its propagator at all momentum scales is characterized by this spectral function. In QCD, the infrared enhancement of the strong coupling leads to nonperturbative spectral-weight accumulation in quark and gluon spectral functions at low energies, a phenomenon indicated by lattice QCD and Dyson–Schwinger equation calculations [13, 14]. In electroweak theory, gauge and Yukawa couplings determine the positions of mass thresholds, while mixing matrices further fix the coupling strengths of different mass eigenstates within the spectral function. The coupling constants of the fundamental interactions determine the structure of each polarization channel in the spectral function and its dependence on mass parameters. The spectral function therefore carries the complete information of the interaction on the two-point propagator.

Following this line of reasoning, could the gravitational interaction likewise be attributed to a modulation of the energy–momentum tensor vacuum fluctuation spectral function under certain conditions? If a massless spin-2 pole were to appear in the spectral function, the most direct interpretation would be the existence of a fundamental massless spin-2 field, just as the δ -peak in the photon spectral function corresponds to the fundamental photon field. However, the Weinberg–Witten theorem rules out this possibility. Jenkins showed that this theorem requires any emergent gravity theory to either lack a covariant energy–momentum tensor or possess a non-relativistic graviton [15]. In a Poincaré-invariant quantum field theory, no Lorentz-covariant, conserved, spin-2 fundamental matrix element exists that could couple to an elementary massless spin-2 particle between single-particle states [16, 17]. Should such a pole appear, it cannot correspond to an elementary particle.

The Weinberg–Witten theorem thus presents a sharp theoretical choice. String theory chooses to negate the point-particle assumption, replacing zero-dimensional particles with one-dimensional extended objects. Loop quantum gravity chooses to negate the background-dependent assumption, quantizing spacetime itself non-perturbatively, one

consequence of which is discreteness at the microscopic scale. The holographic principle chooses to accept that gravity is not fundamental, recasting it as an emergent description of a dual boundary field theory.

Can it signal a composite collective excitation mode? This points to the possibility of gravity as an emergent phenomenon. Like the Casimir effect or Hawking radiation, it is a physical effect produced under specific conditions by the collective statistical behavior of microscopic degrees of freedom. In the Casimir effect, the attractive force between two plates does not arise from a fundamental Casimir force. It is the macroscopic manifestation of the vacuum fluctuation spectral function after modulation by boundary conditions. Similarly, if gravity can emerge from the statistical behavior of energy–momentum tensor fluctuation spectral functions, it would be an effective manifestation of vacuum fluctuation collective effects, as opposed to an independent fundamental force in nature.

This non-fundamental yet emergent picture has a classic analogue in condensed matter physics, the phonon [18–20]. In a crystal, the phonon is not a primitive excitation in the fundamental Lagrangian. It is the quantized manifestation of collective atomic vibrational modes, whose dispersion relation, polarization structure, and effective action are entirely determined by the statistical correlations of microscopic interatomic interactions [21]. There is no need to pre-install a phonon field in the fundamental description. The crystal lattice itself provides the background structure required for phonon emergence. Analogously, if the massless spin-2 excitation is emergent, its background is not a crystal lattice, but the spatial correlation structure of vacuum fluctuations in quantum field theory. From this viewpoint, curved spacetime can be understood as an effective geometric description of the inhomogeneity of such spatial correlations [22, 23]. A massless spin-2 mode, should it emerge, can be understood as the collective excitation of energy–momentum tensor fluctuations on this background. Of course, this analogy has a fundamental limitation. Phonons exist against a fixed crystal lattice background, whereas one of the central challenges of emergent gravity theory is precisely to explain how the spacetime background itself arises from non-geometric degrees of freedom [24]. Nevertheless, it provides a powerful conceptual template. An excitation mode absent at the fundamental level can emerge at low energies through the collective behavior of microscopic degrees of freedom. Its low-energy dynamics are universal, independent of microscopic details.

Temperature, in statistical physics, is another instance of the same logic. In Einstein’s analysis of Brownian motion [25], the irregular motion of a pollen grain is not explained by introducing a new fundamental force. Instead, the collective effect of countless molecular collisions is encoded as a random force, whose statistical properties are linked to viscous drag through the fluctuation–dissipation theorem. Temperature emerges as the macroscopic parameter connecting the two, even though it has no definition at the level of a single molecule. Temperature is not a molecular force pre-installed in the fundamental Lagrangian. It is an effective description of microscopic statistical behavior after coarse-graining.

The present paper applies this logic to gravity. It takes as its sole a priori object the spectral function of the energy–momentum tensor commutator in flat-spacetime quantum field theory, rather than curved spacetime or a fundamental gravitational field. Through

coarse-graining in momentum space and spectral representation, an order-parameter field is constructed. The irreversibility of the coarse-graining process promotes the coarse-graining scale to a dynamical order parameter. Its Langevin dynamics is governed by two renormalization-group fixed points, an ultraviolet repeller at the Planck scale and an infrared attractor at the Hubble scale [1, 26, 27].

The core question we investigate is whether the spin-2 spectral density can develop an isolated zero-momentum pole. The existence or absence of this pole determines whether the order-parameter field can support long-range spin-2 propagation. If the pole is present, Weinberg’s low-energy theorem [1, 28] fixes the infrared self-coupling to the Einstein–Hilbert form. The equivalence principle then follows from the Lorentz structure of the massless spin-2 pole and from the uniqueness of the energy–momentum tensor as the universal current to which matter can couple. Newton’s constant is read from the pole residue at the emergence scale, while the infrared fixed point supplies the natural curvature scale of the effective description. Because the spectral function is defined from the commutator, additive static vacuum-energy terms are absent from the dynamical spin-2 diagnostic. They are not the source of the pole.

The present work follows the spirit of Sakharov’s induced gravity [29, 30], Jacobson’s thermodynamic gravity [31], and Verlinde’s entropic force proposal [32]. He et al. recently constructed gravity by deforming the energy–momentum tensor non-locally, a top-down modification of the operator itself [33]. The present paper provides a bottom-up mechanism. Gravity is not regarded as a fundamental interaction; it is tested as a possible statistical phase of the energy–momentum tensor spectral function under coarse-graining. In contrast to these works, the present paper places the dynamical mechanism in coarse-graining criticality and takes the energy–momentum tensor spectral function as the fundamental object of the theory, as opposed to a fundamental metric or a thermodynamic variable. In this picture, gravity stands to the energy–momentum tensor as temperature stands to molecular motion. Throughout this paper, the term emergence refers specifically to a statistical phase transition in the spectral properties of vacuum fluctuations. Through coarse-graining, the spectral weight of the energy–momentum tensor may reorganize, generating long-range correlations. If this reorganization produces the isolated spin-2 pole, the macroscopic state is described at low energies by general relativity. No new fundamental entity is introduced in this process.

The result is therefore a criterion, not a premise that gravity must emerge in every quantum field theory. The construction starts with the renormalized stress-tensor commutator, defines a coarse-grained spin-2 spectral density, and asks whether its infrared limit contains an isolated zero-momentum pole. Pole formation is a nonperturbative dynamical question. If such a pole is absent, the construction yields a coarse-grained statistical tensor field but not long-range gravity. If the pole is present with positive residue, the low-energy consequences follow from standard spectral representation and soft-graviton arguments. This is the sense in which the proposed mechanism is falsifiable.

2 Spectral Construction of the Coarse-Grained Field

2.1 Overview of the Argument

The starting point is the spin-2 spectral density $\rho^{(2)}(p^2)$ of the energy-momentum tensor commutator. Additive static vacuum terms drop out of the commutator and therefore do not contribute to the propagating spin-2 spectral weight. To isolate the contributions of fluctuations at different length scales, we separate the spectral density into fast and slow modes. We introduce an exponential window function $W_\sigma(p) = e^{-p^2\sigma^2/2}$ to separate fast modes ($p \gtrsim 1/\sigma$) from slow modes ($p \lesssim 1/\sigma$), obtaining the coarse-grained spectral density $\rho_{\text{cg}}(p^2; \sigma) = W_\sigma(p)\rho^{(2)}(p^2)$. This step is irreversible. Discarding fast modes severs their entanglement with the slow modes. This drives the slow-mode reduced density matrix from a pure state to a mixed state, increasing its von Neumann entropy. The coarse-graining scale σ is the central control parameter of the theory. Its inverse $k = 1/\sigma$ is the momentum cutoff. The promotion of σ from a technical parameter to a dynamical order parameter is the subject of §3.1. The physical basis for this promotion is that σ controls how many fast modes have been integrated out, and therefore how much information has been lost. The irreversibility of this information loss generates a net drift in σ , which, together with the stochastic back-reaction of the discarded fast modes, constitutes its Langevin dynamics. For this overview, it suffices that σ inherits its dynamics from the irreversibility of the fast-mode integration. The dynamical emergence of tensor fields has also been investigated in composite gravity models using FRG methods [34].

Discarding the fast modes in one momentum shell produces two independent physical consequences. The first is that the fast modes modify the effective action of the slow modes, i.e. energy transfer. The second is that microscopic information is permanently lost, i.e. entropy production. Both are encoded in the same functional renormalization group trace density. For the exponential regulator $R_k(z) = z/(e^{z/k^2} - 1)$, the energy-transfer and entropy-production kernels are respectively

$$I_E = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k}{(z + R_k + m^2)^2}, \quad I_S = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k}{z + R_k + m^2}.$$

Summing over all matter fields yields the total rates $dE/d\ln k$ and $dS/d\ln k$.

The ratio of entropy production to energy transfer defines an effective temperature:

$$\frac{1}{k_B T_{\text{eff}}(k)} \equiv \frac{dS/d\ln k}{dE/d\ln k}.$$

Evaluating these kernels in the free-field massless zero-mode limit (see Appendix D) gives

$$k_B T_{\text{eff}}(k) = \frac{\hbar c}{2\pi} k = \frac{\hbar c}{2\pi\sigma}.$$

In this limit the temperature is fixed by the same coarse-graining kernels that define the spectral flow. It is not imported from the Unruh effect.

The same trace density that governs energy transfer and entropy production has a third consequence. It dictates how spectral weight is redistributed among the surviving

modes. The integrated-out fast modes carry spectral weight that must be absorbed by the slow modes. The FRG kernel that encodes their energy and entropy contributions also encodes this spectral weight. The coarse-graining flow equation for the spin-2 spectral density is

$$\frac{d}{d\sigma}\rho(\mu^2;\sigma) = -\mu^2\sigma\rho(\mu^2;\sigma) + \Delta_\sigma(\mu^2).$$

The first term describes dilution. Coarse-graining exponentially suppresses spectral weight at all non-zero momenta. The second term describes injection. The integrated-out fast modes deposit their spectral weight into the surviving slow modes. The detailed Feynman-diagram structure of this injection term is presented in Appendix E.

At a fixed point of the coarse-graining flow, the spectral density must be stationary, $d\rho/d\sigma = 0$. Assume the spectral density is smooth at zero momentum, $\rho \sim \mu^4$. Then the dilution term starts at $O(\mu^6)$, which requires the injection term's coefficient at $O(\mu^4)$ to vanish. But the injection term arises from the action of the energy-momentum tensor operator on the fast-mode shell. The energy-momentum tensor is the conserved Noether current of spacetime translations. It has non-zero matrix elements between states carrying energy and momentum, and its zero-momentum transfer channel does not decouple from the fast-mode shell. This produces the obstruction to a smooth fixed point discussed in § 4.2.

Within this spectral-flow class, a stationary configuration cannot remain smooth at $\mu^2 = 0$. The possible non-smooth behaviours of the propagating spectral density are constrained by two structural conditions. The first is positivity, $\rho(\mu^2) \geq 0$, guaranteed by unitarity. The second is the superconvergence sum rule

$$\int_0^\infty d\mu^2 \mu^2 \rho^{(2)}(\mu^2) = 0,$$

which follows from the Ward identity after local contact terms have been separated and is imposed on the subtracted nonlocal spin-2 density. Power-law divergences violate the sum rule. Logarithmic singularities destroy locality. Branch cuts describe threshold continua rather than isolated long-range propagation. The minimal singular configuration compatible with positivity, the sum rule, and locality is an isolated δ -function at $\mu^2 = 0$:

$$\rho^{(2)}(\mu^2) = Z\delta(\mu^2) + \rho_{\text{cont}}(\mu^2),$$

where $Z > 0$ is the pole residue and ρ_{cont} is a non-singular continuum.

Because the energy-momentum tensor is a conserved Noether current, its fluctuations are constrained by the Ward identity. The fluctuations of ordinary composite operators may be diluted away by coarse-graining. The spin-2 stress-tensor weight, by contrast, is tied to the conserved translation current. The superconvergence sum rule restricts how this weight can be redistributed. This is the physical reason why an isolated zero-momentum pole is the natural candidate for a long-range fixed-point component.

Once the spin-2 spectral density possesses an isolated pole at zero momentum, the transverse-traceless part of the coarse-grained field acquires a massless propagator (Appendix C). Weinberg's low-energy theorem fixes the nonlinear self-coupling uniquely to the

Einstein–Hilbert action. This theorem applies to any massless spin-2 mode, irrespective of its microscopic origin. It requires only the existence of the pole, not that the field be fundamental. Newton’s constant is identified with the inverse pole residue, $G = 1/(8\pi Z)$. The infrared fixed-point scale gives the natural curvature scale, $\Lambda \sim 1/\sigma_c^2 = (H_0/c)^2$.

The effective temperature enters the noise correlator through the fluctuation–dissipation theorem. When the coarse-graining scale is promoted to a dynamical order parameter in §3.1, this temperature relates the fast-mode quantum fluctuations to the resulting Langevin dynamics. Where analogies are invoked later, such as the ferromagnet in §2.4 or Brownian motion in §3.3, they serve only as conceptual guides. The technical arguments rest on the QFT structure itself.

The same coarse-graining operation gives rise to the effective temperature. Through the spectral pole, it also gives rise to the massless spin-2 mode. The temperature closes the Langevin dynamics of the order parameter. The low-energy dynamics of this mode is identified by Weinberg’s theorem as general relativity.

2.2 Spectral Representation of the Energy–Momentum Tensor

Gravity couples to matter universally in general relativity, with all matter fields couple to the metric through the same energy–momentum tensor $T_{\mu\nu}$, as encoded in the Einstein equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. The nonlinearity of general relativity reflects a deeper fact, that the gravitational field itself carries energy–momentum, so the energy–momentum tensor couples to itself. This is the strong equivalence principle, the statement that gravitational energy is a source of gravity. It is not an additional assumption but a direct consequence of the self-coupling encoded in the Einstein–Hilbert action. $T_{\mu\nu}$ is both the source of gravity and the carrier of the gravitational field. This dual role makes it the natural starting point for any attempt to go beyond the standard geometric picture.

If gravity is emergent, not fundamental, the energy–momentum tensor, not the metric, should be the basic object of the theory, and the metric should be emergent. In the standard geometric picture, the metric is the fundamental field while $T_{\mu\nu}$ is derived from it. If the picture is reversed, $T_{\mu\nu}$, defined on a flat background, would be fundamental, and the effective metric would be derived from its fluctuation spectrum. As the conserved Noether current of spacetime translations, the energy–momentum tensor is the only current in flat-spacetime quantum field theory that carries spin-2 information. Its vacuum fluctuations, their distribution across momentum scales, and their self-interactions determine whether long-range, gravity-like correlations can arise at low energies.

The spectral representation of the two-point correlator of the energy–momentum tensor is needed to characterize this distribution. The spectral function $\rho(\mu^2)$ encodes the weight with which intermediate states of invariant mass μ contribute to the two-point correlator of the energy–momentum tensor fluctuations.

In its standard form, the Källén-Lehmann representation expresses the full interacting propagator of a field operator as a weighted integral over free propagators,

$$G(p^2) = \int_0^\infty d\mu^2 \frac{\rho(\mu^2)}{p^2 - \mu^2 + i\epsilon},$$

where $\rho(\mu^2)$ is the spectral function of that operator. This relation is exact. It follows from Lorentz invariance, unitarity, and the spectral condition, not from any expansion in a small parameter.

In a free theory, the spectral function reduces to a single δ -function at the particle mass. In an interacting theory, the interaction shifts this peak, broadens it into a continuum above multi-particle thresholds, and introduces new peaks for bound states. The spectral function captures, in full, how the interaction modifies propagation.

The structure of the spectral function tells us what a given field can propagate. An isolated δ -function peak signals a stable single-particle state. The propagator has a pole, and the force it mediates falls off exponentially at large distances. A continuum above a threshold signals the opening of multi-particle channels. The behaviour at zero momentum determines whether the force is long-range. A massless pole produces a $1/r$ potential. The spectral function is a diagnostic tool. Its pole structure dictates the force law.

The spectral function provides a complete, non-perturbative characterization of how the interaction modifies the propagation of the operator’s fluctuations.

The spectral function serves two purposes. It is a diagnostic tool. It tells us which particles exist and which can propagate. It is also a bridge. It connects the microscopic fluctuations of quantum fields to the macroscopic forces they generate. The spectral pole criterion advanced in this paper rests on this dual role. In this language, the emergence of gravity becomes a concrete question. Does the spin-2 spectral density of the energy–momentum tensor develop a massless pole?

Using the vacuum expectation value $\langle \hat{T}_{\mu\nu} \rangle$ as a starting point would be problematic. The divergent vacuum constant proportional to $\eta_{\mu\nu}$ would, in a geometric interpretation, lead to a cosmological constant catastrophe. This paper takes the vacuum matrix element of the energy–momentum tensor commutator as the fundamental object. The commutator eliminates any constant term that commutes with all operators.

The fundamental degrees of freedom are fluctuations, not static backgrounds. The commutator selects precisely those degrees of freedom. Only operators that do not commute with $\hat{T}_{\mu\nu}$ contribute to the spectral function. These are the genuine dynamical fluctuations. The commutator in Eq. (2.2) removes any vacuum-energy contribution, since $E_0\eta_{\mu\nu}$ commutes with $\hat{T}_{\mu\nu}$. Throughout this paper, “propagation” refers to the dynamical propagation of on-shell degrees of freedom, the process by which a massless pole in the spectral function generates a long-range force. It is distinct from the instantaneous transmission of a static force. This principle, encoded at the kinematic level by the commutator, is more than a technical observation. The spectral function is free of vacuum energy contributions not because of a fine-tuned cancellation, but because the formalism restricts to the fluctuating sector. For fundamental interactions, the principle holds in the dynamical sense but is largely tautological, since fundamental particles fluctuate and propagate by definition. When gravity is not a fundamental interaction, the principle becomes non-trivial. Gravity is not pre-installed in the microscopic action; it is selected, if at all, by the infrared pole structure of the stress-tensor spectral density. Whether the energy–momentum tensor fluctuations organise themselves into a propagating spin-2 mode is precisely the question that the spectral pole criterion answers. The commutator, by restricting to the fluctuat-

ing sector, ensures that only those degrees of freedom capable of propagation enter the criterion.

The microscopic theory whose energy–momentum tensor fluctuations we analyse is the Standard Model, defined on four-dimensional flat Euclidean space $\mathbb{R}^4$ with metric $\eta_{\mu\nu}$. Its fields are the Higgs scalar ϕ , quark and lepton spinor fields ψ_f , and gauge vector fields A_μ^a . The action is denoted by S_{micro} , and the field operators satisfy the canonical commutation or anticommutation relations and spectral conditions. The Standard Model contains no fundamental spin-2 field. Any spin-2 propagating degrees of freedom that appear in low-energy physics are emergent collective excitations, not fundamental fields pre-installed in the action. It is the defining condition for gravity to be emergent. The absence of fundamental spin-2 degrees of freedom establishes the prerequisite for the question “can gravity emerge?” to be physically meaningful.

The energy–momentum tensor operator is defined through the functional derivative of the action with respect to a background metric [1, 35]:

$$\hat{T}^{\mu\nu}(x) = \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{micro}}}{\delta g_{\mu\nu}} \Big|_{g_{\mu\nu}=\eta_{\mu\nu}}, \quad (2.1)$$

This operator is symmetric and covariantly conserved, $\partial_\mu \hat{T}^{\mu\nu} = 0$, with mass dimension 4. On a flat background, this definition is equivalent to the canonical form $\hat{T}^{\mu\nu} = \sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \partial^\nu \varphi_a - \eta^{\mu\nu} \mathcal{L}$, where the index a runs over all microscopic field components. The functional derivative definition is advantageous for two reasons. First, it makes explicit how the energy–momentum tensor couples to a background metric. This coupling is a necessary ingredient for the emergence of an effective metric $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ in the low-energy theory. Second, it defines $\hat{T}_{\mu\nu}$ without ever invoking a dynamical metric. The background metric $\eta_{\mu\nu}$ is a fixed mathematical structure, and the functional derivative is evaluated at $g_{\mu\nu} = \eta_{\mu\nu}$. The definition provides the logical link between the flat-spacetime microscopic theory and the effective description in terms of an emergent metric, without presupposing that the emergent metric is curved. The background metric $\eta_{\mu\nu}$ here is the fixed flat spacetime on which the quantum field theory is defined; it is not a dynamical gravitational field. In Euclidean signature, $\eta_{\mu\nu}$ coincides with $\delta_{\mu\nu}$. In this picture, no fundamental curved metric is introduced.

In the following, $\hat{T}_{\mu\nu}$ denotes the renormalized symmetric stress tensor. Its two-point function may contain local contact terms and improvement-term ambiguities. These terms are polynomial in momentum space and do not carry propagating spectral weight. They are separated from the nonlocal part before the spin decomposition is applied. The symbol $\rho^{(2)}$ is reserved for the transverse-traceless discontinuity of this nonlocal part. With this convention, local counterterms can change subtraction constants and spin-0 contact pieces, but not the existence or residue sign of an isolated spin-2 pole at $p^2 = 0$.

The spectral function of the energy–momentum tensor commutator is defined as [36]

$$\rho^{\mu\nu\rho\sigma}(x-y) \equiv \langle 0 | [\hat{T}^{\mu\nu}(x), \hat{T}^{\rho\sigma}(y)] | 0 \rangle. \quad (2.2)$$

Its Fourier transform to momentum space reads

$$\tilde{\rho}^{\mu\nu\rho\sigma}(p) = \int d^4x e^{ip \cdot x} \rho^{\mu\nu\rho\sigma}(x). \quad (2.3)$$

The conservation condition $\partial_\mu \hat{T}^{\mu\nu} = 0$ imposes the transversality constraint $p_\mu \tilde{\rho}^{\mu\nu\rho\sigma} = 0$ in momentum space. The irreducible representations of a symmetric conserved rank-2 tensor contain only spin-2 and spin-0 sectors; the spin-1 part is eliminated by the transversality condition [35]. Introduce the transverse projection operator

$$\theta^{\mu\nu} = \eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}, \quad (2.4)$$

and the spin projection operators

$$P_{\mu\nu\rho\sigma}^{\text{TT}}(p) = \frac{1}{2}(\theta^{\mu\rho}\theta^{\nu\sigma} + \theta^{\mu\sigma}\theta^{\nu\rho}) - \frac{1}{3}\theta^{\mu\nu}\theta^{\rho\sigma}, \quad P_{\mu\nu\rho\sigma}^{\text{S}}(p) = \frac{1}{3}\theta^{\mu\nu}\theta^{\rho\sigma}. \quad (2.5)$$

These operators satisfy idempotence and completeness. The spectral function decomposes into two independent scalar spectral densities [1, 35]:

$$\tilde{\rho}^{\mu\nu\rho\sigma}(p) = \rho^{(2)}(p^2) P_{\mu\nu\rho\sigma}^{\text{TT}}(p) + \rho^{(0)}(p^2) P_{\mu\nu\rho\sigma}^{\text{S}}(p). \quad (2.6)$$

When this decomposition is evaluated in the Standard Model, the result is as follows. In the perturbative expansion, the spin-2 spectral density of massive scalar and spinor fields vanishes identically below the threshold $p^2 < 4m^2$. The spectral densities of massless fields are all proportional to p^4 . The total spin-2 spectral density satisfies, in the infrared limit,

$$\rho^{(2)}(p^2) \sim p^4 \quad (p^2 \rightarrow 0), \quad (2.7)$$

containing no isolated $\delta(p^2)$ pole. This conclusion holds to all orders in perturbation theory.

In the non-perturbative regime, the spectral density of the Standard Model (e.g. in the strongly coupled QCD sector) is currently an open problem. The existence of a pole will determine whether the macroscopic field $\Phi_{\mu\nu}$ can support long-range propagation.

To address this question, we need a framework that tracks how the spectral density evolves from its perturbative form at high energies to its non-perturbative structure at low energies. Coarse-graining is such a framework. By integrating out short-wavelength modes step by step, it reveals how spectral weight is redistributed across momentum scales.

In coarse-graining the quantum field system, we integrate out the fast modes of all fundamental fields. The energy-momentum tensor, as the conserved Noether current of spacetime translations, is the natural object that records how the discarded energy, momentum, and stress redistribute among the surviving slow modes. Its spectral function is the diagnostic tool that reveals whether these fluctuations can organise themselves into a propagating collective mode. The coarse-graining operation is applied directly to this spectral function, a procedure made possible by the Källén-Lehmann representation that encodes the complete two-point information of the energy-momentum tensor fluctuations.

2.3 Coarse-Graining of the Spectral Density

Vacuum fluctuations exist on all momentum scales, but fluctuations at different scales contribute differently to macroscopic physics. Short-wavelength ultraviolet fluctuations manifest as local noise, while long-wavelength infrared fluctuations produce long-range correlations. To separate these two contributions systematically, we introduce a coarse-graining scale $\sigma > 0$, equivalent to a momentum cutoff $k = 1/\sigma$. At scale σ , fast modes with momentum $k \gtrsim 1/\sigma$ are integrated out, and their effects on the slow modes are encoded in the effective dynamics. Slow modes with momentum $k \lesssim 1/\sigma$ are retained.

The coarse-graining operation is implemented directly at the level of the spectral density, using an exponential window function [27, 37]:

$$W_\sigma(p) = e^{-p^2\sigma^2/2}. \quad (2.8)$$

The exponential window is chosen from both physical and technical considerations. Physically, coarse-graining should separate fast and slow modes as impartially as possible. It should introduce no oscillations, respect the positivity of the spectral density, and favor no momentum direction. The exponential function is a smooth cutoff satisfying these requirements. In coordinate space it is Gaussian, consistent with the Gaussian statistics of fast-mode noise. The Gaussian distribution has maximal entropy for a given variance. This is the natural expression of impartiality. For technical convenience, $W_\sigma(p) > 0$ holds for all momenta and depends only on p^2 , which ensures positivity and isotropy of the coarse-grained spectral density. The exponential window factors with the exponential form of propagators, allowing explicit Gaussian integration that simplifies the derivation of evolution equations and heat kernel expansions in later work. The choice of window function affects numerical coefficients, not the scaling behaviour. The effective temperature always satisfies $T_{\text{eff}} \propto 1/\sigma$. The qualitative conclusions of this paper do not depend on the specific regulator.

The coarse-grained spectral density is defined as

$$\rho_{cg}(p^2; \sigma) = W_\sigma(p)\rho(p^2). \quad (2.9)$$

The multiplication by $W_\sigma(p)$ suppresses the contribution of modes with wavelength shorter than σ to the spectral density. Through the Källén-Lehmann representation, the spectral density determines the coordinate-space correlator of the coarse-grained field. Suppressing short-wavelength modes in the spectral density is equivalent to retaining only the long-wavelength contributions to the field correlator. This procedure avoids an intermediate step of operator truncation. It operates directly on the spectral function, which encodes the complete two-point information of the field fluctuations. Since the constant vacuum energy term commutes with all operators, this cutoff raises no subtraction issue related to vacuum expectation values.

2.4 Spectral Construction of the Coarse-Grained Field

By the Källén-Lehmann spectral representation theorem, the coarse-grained spectral density defines a coarse-grained symmetric rank-2 tensor field $\Phi_{\mu\nu}(x)$ at scale σ , whose two-

point correlator is [35, 36]

$$\langle \Phi_{\mu\nu}(x) \Phi_{\alpha\beta}(y) \rangle_\sigma = \int_0^\infty d\mu^2 \rho^{\text{cg}}(\mu^2; \sigma) [P_{\mu\nu\alpha\beta}^{\text{TT}} + P_{\mu\nu\alpha\beta}^{\text{S}}] \Delta(x - y; \mu^2), \quad (2.10)$$

where $\Delta(x - y; \mu^2)$ is the Euclidean scalar propagator of mass μ . Because the coarse-grained spectral density preserves positivity, Wightman's reconstruction theorem guarantees the existence of a unique Hermitian field operator $\hat{\Phi}_{\mu\nu}$ whose vacuum Wightman function is given by (2.10) [1, 2, 35]. The vacuum expectation value of this operator is identically zero,

$$\langle \Phi_{\mu\nu}(x) \rangle = 0. \quad (2.11)$$

This vanishing holds to all orders in the coarse-graining flow, as the commutator defining the spectral density excludes constant contributions by construction.

For any spectral density, $\Phi_{\mu\nu}$ (a coarse-grained statistical field of energy-momentum tensor fluctuations) exists as a quantum field whose two-point function is fully determined by the coarse-grained spectral density. At the Gaussian level, it is a generalized free field describing vacuum stress fluctuations. Whether its transverse-traceless part $\Phi_{\mu\nu}^{\text{TT}}$ supports massless particle propagation depends on whether $\rho^{(2)}$ develops an isolated pole at $p^2 = 0$. If no such pole exists, the correlator of $\Phi_{\mu\nu}^{\text{TT}}$ decays as a power law at large distances. The coarse-grained field still describes vacuum stress fluctuations, and carries no long-range gravitational interaction. Appendix C demonstrates the technical equivalence of this criterion. If the spectral density contains a pole $\rho^{(2)}(\mu^2) = Z\delta(\mu^2) + \dots$, substituting it into the spectral representation (2.10) yields a pole term Z/p^2 in the momentum-space correlator of the transverse-traceless sector, a massless spin-2 propagator producing a $1/r$ force at macroscopic distances. Coarse-graining criticality, as described in this paper, provides a plausible physical mechanism for the formation of such a pole.

Decompose $\Phi_{\mu\nu}$ into irreducible representations [35],

$$\Phi_{\mu\nu} = \Phi_{\mu\nu}^{\text{TT}} + \frac{1}{4}\eta_{\mu\nu}\Phi, \quad \Phi = \Phi_\lambda^\lambda, \quad (2.12)$$

where $\Phi_{\mu\nu}^{\text{TT}}$ is the transverse-traceless part, with spin-2 and 5 polarization degrees of freedom, and Φ is the trace part, with spin-0 and 1 degree of freedom. The two sectors are independently described by $\rho^{(2)}$ and $\rho^{(0)}$ in the spectral representation (2.10).

$\Phi_{\mu\nu}$ inherits its rank-2 tensor character directly from its microscopic construction. The energy-momentum tensor operator $\hat{T}_{\mu\nu}$ is itself a symmetric conserved rank-2 tensor, carrying the spin-2 representation of the Lorentz group. At each spacetime point, its vacuum fluctuation is not a scalar but a 4×4 symmetric matrix whose independent components undergo quantum fluctuations. The spectral representation (2.10) decomposes the correlator of this tensor operator. The weight of each invariant-mass eigenstate μ^2 under the projection operators P^{TT} and P^{S} is determined by the matrix elements of $\hat{T}_{\mu\nu}$. When coarse-graining selects specific momentum regions, the coarse-grained field $\Phi_{\mu\nu}$ inherits the tensor structure and symmetries of its microscopic source $\hat{T}_{\mu\nu}$. This follows the same

principle by which the magnetization field in a ferromagnet inherits the vector character of individual atomic magnetic moments. The symmetry of the order parameter is determined by the symmetry of the microscopic degrees of freedom.

$\Phi_{\mu\nu}$ is not a fundamental field but a coarse-grained statistical field constructed from the vacuum fluctuation spectral function. Its vacuum expectation value is zero by construction, its two-point function is fully determined by the coarse-grained spectral density, and no independent dynamical parameters are introduced.

The effective action of the coarse-grained field contains a quadratic term precisely determined by the spectral representation, together with higher-order self-coupling vertices generated by integrating out the fast modes. In the weak-field, slowly-varying regime, the higher-order terms are suppressed and the Gaussian approximation holds. This condition is automatically satisfied in the infrared limit $\sigma \rightarrow \sigma_c$. As σ approaches the ultraviolet fixed point σ_0 , the Gaussian approximation for the effective action breaks down and higher-order self-couplings must be included. The statistical distribution of the slow modes, however, remains Gaussian at σ_0 . At this scale the cutoff is extremely small, the window function $W_{\sigma_0} \rightarrow 1$, and all fluctuation modes are retained. Fast-mode fluctuations at distinct spatial points are then approximately statistically independent, and the central limit theorem implies that the initial distribution of the slow modes is Gaussian. This initial condition is self-consistent with the exponential cutoff form adopted above [27, 37].

The coarse-graining scale σ and the coarse-grained field $\Phi_{\mu\nu}$ are mutually dependent. Given σ , we define the coarse-grained spectral density and construct the coarse-grained field. In turn, the statistical fluctuations of the coarse-grained field feed back into the evolution of σ through the integration of fast modes.

3 Order Parameter Dynamics and Effective Temperature

The coarse-graining scale σ is not merely a technical parameter. Its dynamics is organized by two fixed points, an ultraviolet repeller at $\sigma_0 \sim \ell_P$ and an infrared attractor at $\sigma_c \sim c/H_0$.

3.1 The Coarse-Graining Scale as an Order Parameter

Quantum field theory is always defined on a fixed flat background metric $\eta_{\mu\nu}$. This is the mathematical prerequisite for the theory to exist. If gravity is emergent, spacetime curvature can be understood as the effective geometric description, at macroscopic scales, of spatial inhomogeneities in the statistical intensity of vacuum fluctuations.

What emerges is not the metric itself, but the dynamics of the metric. The metric, as an effective geometric description of fluctuation inhomogeneity, is emergent. What is not presupposed is the Einstein–Hilbert action. It emerges from the spectral pole via Weinberg’s theorem. Should the spin-2 spectral density of the energy–momentum tensor acquire an isolated pole at the coarse-graining critical point, its vacuum fluctuations acquire long-range correlations. The transverse-traceless part of the coarse-grained field $\Phi_{\mu\nu}$ constructed in §2.4, denoted here by $h_{\mu\nu}$, describes the order parameter of these long-range correlations. It captures the statistical intensity differences of the energy–momentum tensor fluctuations

at distinct spatial points, the spatial distribution of fluctuation inhomogeneity. When matter fields propagate in this inhomogeneous vacuum background, their effective propagation equations are equivalent to those in a curved spacetime. The curved part $\kappa h_{\mu\nu}$ in the effective metric $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ does not originate from a pre-existing geometric structure. It is the macroscopic statistical signature of the inhomogeneity of energy–momentum tensor fluctuations. There is no a priori curved metric, only the inhomogeneous distribution of energy–momentum tensor fluctuations on a flat background. This is consistent with the Weinberg–Witten theorem, which forbids a fundamental massless spin-2 particle in a Poincaré-invariant quantum field theory [16]. If long-range spin-2 correlations emerge, they must do so as collective excitations. The low-energy dynamics of such excitations are then uniquely fixed by Weinberg’s low-energy theorem [28] to the Einstein–Hilbert form. We return to this in §5. When this inhomogeneity acquires long-range correlations at the critical point, it becomes macroscopically equivalent to the curved spacetime of Einstein gravity.

To describe this inhomogeneity, the coarse-graining scale is promoted to a spatially dependent field $\sigma(x) > 0$, called the local correlation length. In the slowly-varying approximation $|\nabla\sigma| \ll 1$, the local cutoff window reads

$$W_{\sigma(x)}(p) = e^{-p^2\sigma(x)^2/2}. \quad (3.1)$$

The dynamics of $\sigma(x)$ is determined by the competition between two limiting scales. The evolution of $\sigma(x)$ is governed by two fixed points, an ultraviolet fixed point $\sigma_0 \sim \ell_P$, unstable and repulsive, and an infrared fixed point $\sigma_c \sim c/H_0$, stable and attractive. At the starting point of the evolution, $\sigma(x) = \sigma_0$ is a uniform constant and the coarse-graining scale serves only as an external control parameter defining the separation between fast and slow modes. The initially uniform vacuum departs from the unstable ultraviolet fixed point under noise perturbations, flows irreversibly toward the infrared fixed point, and gradually develops spatial inhomogeneity in the process.

They follow from the internal logic of the framework developed here. If the statistical intensity differences of vacuum fluctuations at distinct spatial points constitute the microscopic origin and precondition of curved spacetime, then a macroscopic statistical description of this inhomogeneity presupposes the ability to distinguish different spatial points. The Planck length ℓ_P is the scale at which quantum field theory as an effective field theory loses locality. Below ℓ_P , spacetime coordinates lose operational meaning, two distinct spatial points cannot be distinguished, and a spatial difference of fluctuation intensity cannot be defined. The Planck length is the natural lower bound for macroscopic statistical description. The coarse-graining scale σ cannot be smaller than ℓ_P . In the coarse-graining process, σ_0 is set at the Planck scale, not because external observation enforces it, but because this is the smallest scale at which fluctuation inhomogeneity is definable.

The coarse-graining scale σ measures how many fluctuation modes have been integrated out. The larger σ is, the more fluctuation modes are subsumed into the definition of the coarse-grained field. When σ exceeds a critical value, the retained slow modes no

longer carry enough fluctuation information to maintain statistical correlations between two spatial points. This critical value is set by the largest distance of causal contact in the universe, namely the Hubble horizon c/H_0 . Two points beyond this distance have never entered each other's causal past and cannot be physically correlated. Hence $\sigma_c \sim c/H_0$ is the largest scale at which correlations can exist. Beyond it, macroscopic statistical description loses its object. The associated infrared curvature scale is $\Lambda \sim 1/\sigma_c^2$. In this formulation it is tied to the endpoint of the coarse-graining flow rather than to an additive vacuum-energy constant. The commutator construction of §2.2 removes such constants from the propagating spectral diagnostic.

The two fixed points have a physical interpretation. The scale σ_0 is the natural ultraviolet boundary for the statistical description of fluctuation inhomogeneity. σ_c is the natural infrared boundary. The framework requires only that these two fixed points exist with the correct stability directions, σ_0 repulsive and σ_c attractive. The ratio $\sigma_0/\sigma_c \sim 10^{-61}$ is not derived from the coarse-graining process. It comes from the observed ratio of the Planck length to the Hubble radius, and this ratio sets the dynamic range of the coarse-graining flow.

Each step of the coarse-graining operation discards short-wavelength information. The process is irreversible and defines a direction from ultraviolet to infrared. This irreversibility means that the spatial integral of $\sigma(x)$ tends to grow along the coarse-graining flow, reflecting the net drift from the ultraviolet toward the infrared. It is not conserved and belongs to the category of non-conserved order parameters. Integrating out the fast modes from the path integral yields an effective action for the slow mode σ . Expanding to second order in the slow-mode fields and performing a Hubbard–Stratonovich transformation to introduce an auxiliary noise field, we obtain from the saddle-point condition an equation of motion for σ . In the vicinity of the coarse-graining critical point, the relaxation time of the slow mode is parametrically long compared to its intrinsic oscillation period. In this overdamped limit the inertial term can be neglected and the equation reduces to the relaxational form. The full derivation, including the determination of the dissipation kernel and the noise correlator in terms of the fast-mode spectral function, is given in Appendix A. The resulting Langevin dynamics for the order parameter is that of a non-conserved field, Model A in the Hohenberg–Halperin classification [1, 26]. The prefactor σ/c ensures the correct scaling dimension of the relaxation rate and is consistent with the dissipative time scale obtained in Appendix A:

$$\frac{\partial \sigma}{\partial t} = \frac{\sigma}{c} [D \nabla^2 \sigma + \beta(\sigma) + \eta(t, x)] , \quad (3.2)$$

The logic that leads from coarse-graining to an effective temperature is the same logic that leads from coarse-graining to the spectral criterion for gravity. Both originate in the irreversibility of the fast-mode integration. The effective temperature encodes the statistical effect of the discarded modes on the scalar channel of the trace density. The spectral pole, should it form, encodes their effect on the spin-2 channel. Temperature and gravity appear as two macroscopic aspects of the same microscopic source, the vacuum fluctua-

tions of quantum fields. What the fluctuation–dissipation theorem does for temperature, Weinberg’s low-energy theorem does for gravity.

The diffusion term $D\nabla^2\sigma$ ($D > 0$) erases spatial inhomogeneity, consistent with the principle that unbiased variation drives the system toward a homogeneous state. The deterministic drift term $\beta(\sigma) = -\delta V_{\text{eff}}/\delta\sigma$ is determined by the gradient of an effective potential, driving the system toward minima of V_{eff} . At the fixed points themselves, the drift is expected to be parametrically small compared to the characteristic scale of the effective potential. The sign conditions on $\beta(\sigma)$ at the two endpoints guarantee that the coarse-graining flow is directed irreversibly from the ultraviolet toward the infrared. The noise term $\eta(t, x)$ originates from the back-reaction of fast-mode fluctuations on the slow modes. Its statistical properties are constrained by the fluctuation–dissipation theorem.

The order parameter $\sigma(x)$ is not a fundamental quantum field. It is a coarse-grained collective variable that encodes the statistical properties of the integrated fast modes. Its dynamics, described by the Langevin equation (3.2), is classical in nature because the quantum coherence of the underlying fast modes is lost at each coarse-graining step. The noise term η represents the residual quantum fluctuations that survive the coarse-graining process, and the fluctuation–dissipation theorem guarantees that these fluctuations are consistent with the effective temperature inherited from the discarded modes.

The full non-perturbative form of the effective potential $V_{\text{eff}}(\sigma)$ is encoded in the functional renormalization group equations of the Standard Model on the corresponding background. For the spectral pole criterion, only the endpoint structure and the sign of the drift in their neighbourhoods enter, and these data can be stated without specifying a detailed function.

The order parameter σ flows from the ultraviolet fixed point σ_0 to the infrared fixed point σ_c . The deterministic drift $\beta(\sigma) = -\partial V_{\text{eff}}/\partial\sigma$ must be positive near σ_0 , so that the ultraviolet fixed point repels, and negative near σ_c , so that the infrared fixed point attracts. These sign conditions, together with the parametrically small magnitude of $\beta(\sigma)$ at the fixed points themselves, isolate the two endpoints as the only stationary configurations of the coarse-grained dynamics.

No intermediate zero of $\beta(\sigma)$ is required by the logic of the framework. The spectral pole formation mechanism of §4 does not rely on critical slowing-down at any intermediate scale. The relaxation rate $\gamma(\sigma)$ that appears in the Langevin equation is the dissipative back-reaction of the integrated fast modes, whose microscopic origin and behaviour are discussed in §3.4. Together, $\beta(\sigma)$ and $\gamma(\sigma)$ encode the two facets of the fast-mode integration, the deterministic drift that pushes σ toward the infrared and the dissipative and stochastic effects that accompany the loss of information at each coarse-graining step.

The Langevin drift $\beta(\sigma)$ must be distinguished from the renormalization group scaling flow $k\partial_k\sigma$. Along a self-similar RG trajectory, the scale σ is simply the inverse of the momentum cutoff, $\sigma = 1/k$, and the RG flow obeys $k\partial_k\sigma = -\sigma$. This scaling relation is a property of the Wilsonian coarse-graining procedure, not an additional dynamical equation. The Langevin equation (3.2) governs a different aspect of the physics, the coarse-grained relaxation of spatial inhomogeneities in $\sigma(x)$ toward equilibrium. The two are compatible, both drive the system from ultraviolet to infrared, but they operate at different levels

of description. The framework requires only that the deterministic part of the Langevin dynamics respects the direction set by the underlying RG flow, which is guaranteed by the sign conditions on $\beta(\sigma)$ at the two endpoints.

The qualitative picture does not depend on the specific functional form of $\beta(\sigma)$ within the physical interval. In this picture, gravity is not a fundamental ingredient of the microscopic action. Whether it emerges, whether the energy–momentum tensor fluctuations succeed in organising themselves into a propagating spin-2 mode, is the question the spectral pole criterion answers. Should the pole form, the resulting massless spin-2 excitation, the graviton in this picture, is not a fundamental particle but the quantum of the emergent order parameter $h_{\mu\nu}$. Near σ_0 , the order parameter is driven away from the Planck scale by the combination of a positive drift $\beta(\sigma) > 0$ and a strong stochastic noise inherited from the large effective temperature. As σ increases, the effective temperature drops, the noise weakens, and near σ_c the drift turns negative and guides σ into the infrared attractor. The detailed shape of $\beta(\sigma)$ in the interior of the physical interval does not affect the conclusions of this paper.

3.2 Coupling to Matter

The interaction of the coarse-grained field $\Phi_{\mu\nu}$ with matter fields is fixed by the Lorentz transformation properties of the fields involved. $\Phi_{\mu\nu}$ is a symmetric rank-2 tensor operator. The energy–momentum tensor of matter fields, $T_{\mu\nu}$, is the symmetric conserved rank-2 tensor current of the Standard Model, with the same Lorentz transformation properties. The linear coupling between two symmetric tensors of the same rank is the full index contraction,

$$S_{\text{coupling}} = \int d^4x T^{\mu\nu}(x) \Phi_{\mu\nu}(x). \quad (3.3)$$

In general relativity, the universal coupling of the metric to the matter energy–momentum tensor is a requirement of the equivalence principle, built into the structure of the Lagrangian. In the present construction the same universality arises from the available spin-2 current. Every matter field that possesses energy–momentum contributes to the same total stress tensor $T_{\mu\nu}$, and the coarse-grained field $\Phi_{\mu\nu}$, being a symmetric rank-2 tensor, couples linearly to this current through the full index contraction. Up to improvement terms and higher-derivative operators, no independent linear Lorentz-invariant coupling exists in the spin-2 channel. This gives the weak equivalence principle at leading long distances.

The Einstein equivalence principle follows with equal necessity. Because $\Phi_{\mu\nu}$ is the only symmetric rank-2 field available to encode the statistical inhomogeneity of vacuum fluctuations, the effective geometry $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ is unique. Matter fields propagating in this effective geometry obey equations that are locally indistinguishable from those in a genuinely curved spacetime. The local Lorentz invariance of the underlying flat-spacetime quantum field theory guarantees that, in a freely falling frame, these equations reduce to their standard special-relativistic form. The equivalence principle follows from three facts within this framework: the Lorentz structure of the Standard Model, the uniqueness of

$T_{\mu\nu}$ as the spin-2 current, and the identification of $h_{\mu\nu}$ as the only available macroscopic spin-2 field.

The strong equivalence principle follows from the same logic at one higher order in the pole phase. Gravitational energy then gravitates because the nonlinear Einstein–Hilbert completion is fixed by Weinberg’s low-energy theorem. The nonlinear terms in the Einstein equations become the macroscopic signature of the self-coupling of energy–momentum tensor fluctuations. Thus the equivalence principle is not inserted as a separate postulate; it is inherited from the spin-2 pole, the stress-tensor coupling, and the low-energy consistency conditions.

In the free Gaussian approximation, the external source does not alter the polarization structure or momentum dependence of the correlator. It only produces an overall amplitude rescaling,

$$\langle \Phi_{\mu\nu}(x) \Phi_{\alpha\beta}(y) \rangle_T = A(\sigma) \langle \Phi_{\mu\nu}(x) \Phi_{\alpha\beta}(y) \rangle_0. \quad (3.4)$$

The factor $A(\sigma)$ characterizes how matter sources modulate the coarse-grained field amplitude. By injecting energy–momentum into the vacuum, they alter the fluctuation amplitude of $\Phi_{\mu\nu}$. There is no direct coupling between Φ and σ . Their mutual influence is mediated indirectly through the cutoff window and the noise term of the Langevin equation.

3.3 Entropy Production and Effective Temperature

Every coarse-graining step discards the fast modes in a momentum shell $[\sigma^{-1}, (\sigma + d\sigma)^{-1}]$. Discarding these modes has two independent physical consequences. They shift the effective action of the slow modes, and their microscopic information is lost, increasing the entropy of the slow-mode reduced density matrix. The ratio of these two effects defines the effective temperature.

The fast-mode integration is the coarse-graining analogue of Brownian motion: the discarded modes act as a heat bath determining T_{eff} for the slow modes via the fluctuation–dissipation theorem.

The coarse-graining operation is meaningful only when applied to a system with sufficiently many degrees of freedom. If only a small number of modes were involved, discarding them would produce no well-defined notion of entropy, no thermodynamic limit, and no effective temperature. The vacuum of quantum field theory supplies the infinite-dimensional system needed for coarse-graining to define entropy and temperature. Even in the extreme ultraviolet, where fluctuations at distinct spacetime points become statistically independent, every point still hosts infinitely many field modes. Coarse-graining does not need to wait for statistics to emerge. It operates on a statistical system from the start.

The relevant question is not where statistics begins, but where coarse-graining ceases to apply. This happens when the number of integrated modes becomes too small to sustain a thermodynamic description. In the far infrared, only zero modes survive, the effective temperature approaches zero, and the Langevin equation reduces to classical relaxation. Throughout the entire range of scales that the framework covers, from the Planck scale to

the Hubble scale, the number of modes being coarse-grained remains large enough for the operation to be valid.

This transition from microscopic to macroscopic description is accompanied by a change in the method of analysis. On the microscopic side, fast modes are integrated out mode by mode, and each step is an exact operation in quantum field theory. The FRG trace density captures this process precisely. On the macroscopic side, the surviving slow modes are sufficiently numerous to define thermodynamic quantities such as entropy and temperature. The trace density produced by the microscopic analysis is projected onto the slow-mode subspace, yielding an entropy production rate and an energy transfer rate. Their ratio defines the effective temperature, which then drives the Langevin equation. No external assumptions are introduced in this mapping from quantum fluctuations to thermodynamic variables.

In the FRG framework, the contribution of the integrated fast modes to the effective action is given by the trace density. For the exponential regulator

$$R_k(z) = \frac{z}{e^{z/k^2} - 1}, \quad z = p_0^2 + \lambda_n, \quad (3.5)$$

a mode with eigenvalue λ_n and effective mass m^2 contributes to the energy transfer via the kernel

$$\mathcal{I}_E(\lambda_n, m^2, k) = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k(z)}{(z + R_k(z) + m^2)^2}, \quad (3.6)$$

where $\partial_t \equiv \partial/\partial \ln k$ is the RG time derivative. Summing over the full matter spectrum gives the total energy transfer rate

$$\frac{dE}{d \ln k} = \frac{1}{2} \sum_n d_n \cdot \mathcal{I}_E(\lambda_n, m^2, k), \quad (3.7)$$

with d_n the degeneracy of each mode. This energy transfer rate follows directly from the Wetterich equation. The trace density is the response of the effective action to the RG scale.

The same integrated fast modes carry entanglement entropy before being discarded. The entropy production kernel shares the same mathematical structure as the energy kernel but with a different weight function. At one loop, the entanglement entropy is proportional to the logarithm of the propagator; its RG derivative yields a single power of the denominator:

$$\mathcal{I}_S(\lambda_n, m^2, k) = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k(z)}{z + R_k(z) + m^2}. \quad (3.8)$$

The total entropy production rate is

$$\frac{dS}{d \ln k} = \frac{k_B}{2} \sum_n d_n \cdot \mathcal{I}_S(\lambda_n, m^2, k). \quad (3.9)$$

The effective temperature is defined as the ratio of entropy production to energy transfer:

$$\frac{1}{k_B T_{\text{eff}}(k)} \equiv \frac{dS/d \ln k}{dE/d \ln k}. \quad (3.10)$$

As a direct verification, we compute the two kernels explicitly in the free-field massless zero-mode limit ($\lambda_n = 0, m^2 = 0$).

For the energy kernel, $z = p_0^2$. Substituting the explicit forms of R_k and $\partial_t R_k$ and simplifying algebraically, the integrand reduces to

$$\frac{\partial_t R_k}{(p_0^2 + R_k)^2} = \frac{2}{k^2} e^{-p_0^2/k^2}, \quad (3.11)$$

which integrates to

$$\mathcal{I}_E(0, 0, k) = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{2}{k^2} e^{-p_0^2/k^2} = \frac{1}{\sqrt{\pi}k}. \quad (3.12)$$

For the entropy kernel, the integrand reduces to

$$\frac{\partial_t R_k}{p_0^2 + R_k} = \frac{2p_0^2}{k^2} \frac{1}{e^{p_0^2/k^2} - 1}, \quad (3.13)$$

which integrates to

$$\mathcal{I}_S(0, 0, k) = \frac{k}{\pi} \int_0^{\infty} \frac{2x^2}{e^{x^2} - 1} dx = \frac{k}{2\sqrt{\pi}} \zeta\left(\frac{3}{2}\right), \quad (3.14)$$

where $\zeta(3/2) \approx 2.612$ is the Riemann zeta function. The ratio of the two kernels is

$$\frac{\mathcal{I}_S}{\mathcal{I}_E} = \frac{k^2}{2} \zeta\left(\frac{3}{2}\right) \approx 1.306 k^2. \quad (3.15)$$

This ratio is proportional to k^2 , with the numerical coefficient depending on the regulator choice: the Litim sharp cutoff gives exactly 1, while the exponential window gives approximately 1.306. Such regulator dependence is a natural feature of the FRG. Different cutoff schemes correspond to different coarse-graining prescriptions, but the scaling $T_{\text{eff}} \propto k$ is universal across all schemes.

The effective temperature is determined by the ratio of total entropy production to total energy transfer. In the free-field, massless, zero-mode limit, the phase-space factor $1/(2\pi)$ combines with the fluctuation–dissipation theorem mapping to yield

$$k_B T_{\text{eff}}(k) = \frac{\hbar c}{2\pi} k. \quad (3.16)$$

The π in the coefficient originates from the standard $1/(2\pi)$ normalization of the Fourier transform in quantum field theory, which is universal across all regulator choices; the 2 comes from the specific form of the exponential window function. For the Litim regulator, the corresponding coefficient is $1/\pi$. We adopt the exponential window throughout, because it directly corresponds to the coarse-graining window $W_\sigma(p) = e^{-p^2\sigma^2/2}$ of Section 2.3, ensuring consistency of the cutoff scheme across the entire framework (see also Appendix A, A.5).

With the effective temperature determined, the noise correlator is bound to the dissipation coefficient through the fluctuation–dissipation theorem:

$$\langle \eta(t, x) \eta(t', x') \rangle = 2\gamma(\sigma) k_B T_{\text{eff}} \delta(t - t') \delta^{(3)}(x - x'). \quad (3.17)$$

This is the standard fluctuation–dissipation relation. Applying it to the order parameter σ uses the linear response of the cutoff window to the fast modes. Along a self-similar trajectory the dissipation coefficient and the effective temperature are both determined by the same trace density; their constant ratio is a direct consequence of the fluctuation–dissipation theorem. Away from such a trajectory, nonlinear corrections change the detailed kernel but not the pole criterion.

Inserting the effective temperature $k_B T_{\text{eff}} = \hbar c / (2\pi\sigma)$ and the relaxation rate $\gamma(\sigma)$ from §3.4 into the Langevin equation (3.2) gives the closed Markovian approximation used in this paper. The fast modes set both the dissipation and the noise through the same trace density. The slow modes obey Langevin dynamics in the regime where scale separation justifies the local fluctuation–dissipation form.

The following points clarify the physical nature of T_{eff} .

Origin of the entropy. T_{eff} is a statistical parameter that encodes the ratio of entropy production to energy transfer intrinsic to the coarse-graining operation. Its physical role parallels that of thermodynamic temperature in the fluctuation–dissipation theorem, but its microscopic origin is quantum entanglement instead of thermal motion. At each coarse-graining step, the discarded fast modes are entangled with the retained slow modes. Tracing them out severs this entanglement, driving the reduced density matrix of the slow modes from a pure state to a mixed state. The resulting increase in von Neumann entropy is the physical source of the kernel $\mathcal{I}_S(\lambda_n, m^2, k)$ in equation (3.8). This mechanism is structurally parallel to the information loss that underlies Hawking radiation, but it requires neither an event horizon nor a curved spacetime. The entanglement truncation is effected entirely by the coarse-graining operation in flat spacetime. T_{eff} is not a thermodynamic temperature. There is no external heat bath. The fast modes, before being integrated out, reside in the quantum vacuum state, not a thermal state.

The Unruh temperature corresponds to the same formula with σ replaced by the horizon scale, though the physical mechanisms are distinct. T_{eff} is an intrinsic consequence of the coarse-graining operation in flat-spacetime quantum field theory, independent of accelerated observers or spacetime geometry.

Regime of validity. In the regime $k_B T_{\text{eff}} \gg \hbar\omega$, where ω is the characteristic frequency of the slow mode, the quantum fluctuation–dissipation theorem reduces to its classical form, $\nu(\omega) = 2k_B T_{\text{eff}} \Gamma(\omega)$. This condition is satisfied in the vicinity of the emergence scale, where the slow-mode frequency is parametrically small compared to the effective temperature. Deeper in the infrared, as σ approaches σ_c and $T_{\text{eff}} \rightarrow 0$, the quantum fluctuation–dissipation relation must be restored (see Appendix A, A.5). The central criterion of this framework, whether the spectral pole forms, involves physics in the vicinity of the emergence scale, where the classical form of the fluctuation–dissipation theorem is applicable.

3.4 Relaxation Rate and Noise-Driven Escape from the Ultraviolet

The relaxation rate $\gamma(\sigma)$ was introduced in §3.1 as the dissipative back-reaction of the integrated fast modes on the slow dynamics of σ (see Appendix A, A.4 for the microscopic derivation):

$$\gamma(\sigma) \sim \int_{|k| \sim 1/\sigma} d^3k |V(k)|^2 \rho_{\text{fast}}(k) \delta(\omega_k - \omega_{\text{slow}}). \quad (3.30)$$

The coupling vertex $V(k)$ arises from the dependence of the coarse-graining window $W_\sigma(p) = e^{-p^2\sigma^2/2}$ on the order parameter. When σ fluctuates, the cutoff scale shifts, and modes near the shell boundary move across the cutoff. The fast-mode spectral density $\rho_{\text{fast}}(k)$ is determined by the energy–momentum tensor fluctuations of the Standard Model fields at scale k .

In the ultraviolet, the number of active fast modes is maximal. All Standard Model fields (quarks, leptons, gauge bosons, and the Higgs doublet) contribute to the dissipation kernel. The integral takes its largest value at the ultraviolet fixed point σ_0 . As the coarse-graining flow proceeds toward the infrared, fast-mode degrees of freedom are successively integrated out, the phase space for dissipation shrinks, and $\gamma(\sigma)$ decreases monotonically.

The effective temperature $T_{\text{eff}} = \hbar c/(2\pi\sigma)$ diverges as $\sigma \rightarrow 0$. The noise correlator is tied to the dissipation by the fluctuation–dissipation theorem:

$$\langle \eta(t, \mathbf{x}) \eta(t', \mathbf{x}') \rangle = 2\gamma(\sigma) k_B T_{\text{eff}} \delta(t - t') \delta^{(3)}(\mathbf{x} - \mathbf{x}'). \quad (3.31)$$

Since $\gamma(\sigma_0)$ is finite and T_{eff} diverges, the noise strength is large in the ultraviolet. The escape of the order parameter from the Planck-scale fixed point is noise-driven. As σ increases and T_{eff} drops, the noise weakens. In the infrared, $T_{\text{eff}} \rightarrow 0$ and the Langevin dynamics reduces to classical relaxation toward the attractor at σ_c .

The fluctuation–dissipation theorem remains the bridge between the microscopic fast-mode integration and the macroscopic Langevin dynamics. It does not require $\gamma(\sigma_0)$ to vanish. It requires only that the noise and the dissipation share a common microscopic origin in the same fast-mode trace density, which they do. The irreversibility of the coarse-graining flow is guaranteed by the direction of the deterministic drift $\beta(\sigma)$, which is positive near σ_0 and negative near σ_c , as established in §3.1.

4 Coarse-Graining Criticality and Spectral Poles

4.1 The Condition for Spectral Pole Formation

The emergence of a massless spin-2 excitation, should it occur, parallels the logic that defines temperature from the fluctuation-dissipation relation. Both are macroscopic quantities extracted from the same coarse-graining trace density. Temperature comes from entropy production. The spectral pole comes from the accumulation of spectral weight at zero momentum.

The decrease of T_{eff} with σ reflects the progressive loss of entanglement between the slow modes and the integrated-out fast modes. In the far ultraviolet, where the cutoff σ is small and T_{eff} is large, the spectral function retains its perturbative form $\rho^{(2)}(p^2) \sim p^4$, which is sustained by the nearly uncorrelated character of fluctuations at neighbouring spacetime points. As σ grows and T_{eff} drops, the accumulated entanglement truncation imprints increasingly long-ranged statistical correlations on the slow modes. At some scale

σ_G , the correlation length becomes comparable to the characteristic wavelength of the lowest-lying collective spin-2 excitations of the energy-momentum tensor. The perturbative spectral configuration can no longer be maintained. The system may undergo a reorganization of spectral weight. The scale σ_G is set by the condition that the effective temperature $k_B T_{\text{eff}} = \hbar c / (2\pi\sigma)$ becomes comparable to the characteristic energy scale of these collective excitations. This transition would occur well away from the ultraviolet fixed point, where the relaxation rate has already decreased from its ultraviolet maximum and the noise has weakened, allowing long-range order to develop.

4.2 Obstruction to Smooth Fixed Points

The obstruction to smooth fixed points does not rely on the specific mechanism—spectral weight reorganization or entanglement accumulation—that drives the flow. It follows from the structure of the dilution and injection terms in the spectral flow used here.

The evolution of the spin-2 spectral density under an infinitesimal change of the coarse-graining scale follows from the definition of the coarse-graining operation. Let $\rho^{(2)}(p^2)$ be the spin-2 spectral function before coarse-graining. The coarse-grained spectral density at scale σ is $\rho(p^2; \sigma) = W_\sigma(p) \rho^{(2)}(p^2)$, where $W_\sigma(p) = \exp(-p^2 \sigma^2 / 2)$ is the exponential window function. Differentiating with respect to σ gives

$$\frac{d}{d\sigma} \rho(p^2; \sigma) = \frac{\partial W_\sigma}{\partial \sigma} \rho^{(2)}(p^2) = -p^2 \sigma W_\sigma(p) \rho^{(2)}(p^2) = -p^2 \sigma \rho(p^2; \sigma). \quad (4.1)$$

This is the dilution term. In the full coarse-graining process, the modes in the momentum shell $[\sigma^{-1}, (\sigma + d\sigma)^{-1}]$ are not simply discarded. They are integrated out, and their spectral weight is injected into the remaining slow modes. This injection is described by $\Delta_\sigma(p^2)$, whose contribution arises from the fast-mode propagator and the energy-momentum tensor interaction vertex. Adding the injection term to the dilution contribution yields

$$\frac{d}{d\sigma} \rho(\mu^2; \sigma) = -\mu^2 \sigma \rho(\mu^2; \sigma) + \Delta_\sigma(\mu^2), \quad (4.2)$$

where $\mu = |p|$ in the Euclidean convention.

At the ultraviolet fixed point σ_0 , all modes are retained ($W_{\sigma_0} \rightarrow 1$), and the injection Δ_{σ_0} is generated entirely by the bare interaction vertices. Away from σ_0 , the injection depends on the spectral density of the already-integrated modes.

Equation (4.2) is the Wilsonian spectral flow used in this paper, with the injection term Δ_σ playing the role of the regulator-dependent trace density in the standard FRG flow for the effective action. The two flows act on different objects but encode the same fast-mode integration. At the critical point, where the order parameter evolves slowly, the spectral density is expected to settle into a configuration that no longer drifts under coarse-graining. The fixed-point condition $d\rho/d\sigma = 0$ states that the configuration is stationary under this flow. A contradiction argument then excludes a smooth stationary configuration within this flow class.

Consider a putative scale σ_* at which the spectral density is assumed to be stationary. The fixed-point condition $d\rho/d\sigma = 0$ implies a balance between dilution and injection.

Assume the spectral function at this scale is smooth at zero momentum and has no isolated pole, that is, $\rho(\mu^2; \sigma_*) = a\mu^4 + O(\mu^6)$. Substituting the smooth expansion into the flow equation gives

$$-\sigma_* a \mu^6 + \Delta_\sigma(\mu^2) = 0 + O(\mu^8). \quad (4.3)$$

The dilution term starts at $O(\mu^6)$. For Eq. (4.3) to hold at $O(\mu^4)$, the coefficient of μ^4 in $\Delta_\sigma(\mu^2)$ must vanish exactly.

The fast-mode shell generates the injection term $\Delta_\sigma(\mu^2)$. Expanding near $\mu^2 = 0$,

$$\Delta_\sigma(\mu^2) = \frac{1}{\sigma^4} [A_0 + A_1 \mu^2 \sigma^2 + A_2 \mu^4 \sigma^4 + \dots], \quad (4.4)$$

where the coefficients A_n are set by the fields in the fast-mode shell. The coefficient A_2 at $O(\mu^4)$ is nonzero in the flow considered here. Two independent arguments support this conclusion.

In the heat kernel expansion the Seeley–DeWitt coefficient a_2 receives contributions from every active matter field [38]. No symmetry of the stress-tensor channel cancels this coefficient. Appendix E reaches the same conclusion by classifying the Feynman diagrams that contribute to Δ_σ . The non-vanishing of the injection term follows from the non-vanishing of the energy–momentum tensor matrix elements at zero momentum transfer. For a conserved Noether current these matrix elements cannot vanish. The two arguments are independent and point to the same result.

Hence $A_2 \neq 0$ at σ_* . This contradicts the fixed-point condition (4.3), which requires the $O(\mu^4)$ coefficient of $\Delta_\sigma(\mu^2)$ to vanish. Within the spectral-flow class defined by (4.2), no smooth spectral function can be a fixed point under these assumptions.

4.3 The δ -Function as the Natural Candidate

If no smooth spectral function can be a fixed point within this flow class, then any stationary configuration of that flow must be nonsmooth at $\mu^2 = 0$. The constraints of positivity, unitarity, and the superconvergence sum rule do not uniquely determine the form of this nonsmoothness. However, among the possible candidates, the isolated δ -function at $\mu^2 = 0$ emerges as the minimal physically consistent choice, for the following reasons. The possible nonsmooth behaviors of the propagating spectral density are constrained by two properties.

$$\int_0^\infty d\mu^2 \mu^2 \rho^{(2)}(\mu^2) = 0, \quad (4.5)$$

The sum rule is understood for the renormalized nonlocal spin-2 density after all local polynomial contact terms have been separated, as in § 2.2. It is the Ward-identity constraint on the propagating transverse-traceless density. Positivity and this sum rule restrict nonsmooth behaviours. Power-law divergences violate the sum rule; logarithmic singularities imply nonlocality; branch points correspond to multi-particle thresholds with vanishing phase space near $\mu^2 = 0$. Within the Källén–Lehmann framework, the isolated δ -function at $\mu^2 = 0$ is the natural candidate. A self-consistent computation of the graviton

spectral function in Lorentzian quantum gravity, based on different assumptions, independently finds a massless one-graviton peak [39]. The continuum contribution is phase-space suppressed near threshold.

The above argument thus identifies the δ -function as the natural candidate for the nonsmooth fixed-point configuration at zero momentum. The physical origin of the transition that enables this selection is the statistical emergence of long-range correlations. When the coarse-graining flow develops a maximally symmetric correlation structure, the energy-momentum tensor fluctuations may gain an ordered substrate. The long-wavelength fluctuations may then organise into collective motion on this substrate, whose lowest excitations are naturally massless. This statistical emergence mechanism operates when the coarse-graining flow reaches a regime where the integrated fast modes have imprinted a sufficiently symmetric effective action on the slow modes. It does not require the order parameter to freeze at an intermediate critical point. Whether triggered by the accumulation of entanglement truncation at σ_G or by other aspects of the coarse-graining dynamics, the consequence for the low-energy theory is the same: a massless spin-2 pole appears, and Weinberg’s theorem fixes its infrared dynamics to be Einstein gravity. This is the nonperturbative content tested by the spectral criterion.

5 Low-Energy Effective Theory of Gravity

Given the existence of an isolated pole at zero momentum in the spin-2 spectral density, we discuss the low-energy behavior of the macroscopic field $\Phi_{\mu\nu}$ on a flat background. The following derivation assumes three conditions. First, the total spin-2 spectral density takes the form $\rho^{(2)}(p^2) = Z \delta(p^2) + \rho_{\text{cont}}(p^2)$, with $Z > 0$ the pole residue and $\rho_{\text{cont}}(p^2)$ a smooth continuum part. Second, coarse-graining has been advanced to the infrared scale $\sigma \approx \sigma_c$. Third, the metric perturbation amplitude is small.

Remark 5.1 (Relation to the Weinberg–Witten theorem). The pole assumed in this section is not an elementary massless spin-2 particle added to the microscopic flat-spacetime theory. It is a pole in the coarse-grained stress-tensor spectral density. If one tried to interpret it as a fundamental asymptotic particle carrying a Lorentz-covariant conserved stress-tensor charge in the microscopic Hilbert space, the Weinberg–Witten obstruction would apply. The present construction instead treats the pole as a collective infrared mode of the renormalized stress-tensor fluctuations. The theorem is therefore a boundary condition on the interpretation of the pole: the mode may be emergent, but it cannot be fundamental.

5.1 Pole and Free Propagator

Under the pole hypothesis, the momentum-space correlation function of the spin-2 sector in the spectral representation (2.10) contains an isolated pole term [35, 36],

$$\langle \Phi_{\mu\nu}^{\text{TT}}(p) \Phi_{\alpha\beta}^{\text{TT}}(-p) \rangle \supset \frac{Z}{p^2} \mathcal{P}_{\mu\nu\alpha\beta}^{\text{TT}}(p). \quad (5.1)$$

The detailed steps of this derivation are given in Appendix C. In coordinate space, this pole term is equivalent to the Pauli–Fierz action, which describes a free massless spin-2

field propagating on a flat background [1, 35],

$$S_{\text{free}} = \frac{1}{2\kappa^2} \int d^4x h^{\mu\nu} \mathcal{E}_{\mu\nu}{}^{\rho\sigma} h_{\rho\sigma}, \quad (5.2)$$

where $h_{\mu\nu} \equiv \kappa \Phi_{\mu\nu}^{\text{TT}}$, $\kappa^2 = 1/Z$, and $\mathcal{E}$ is the linearized Einstein tensor operator. The Newton constant G is read from the pole residue,

$$G = \frac{\kappa^2}{8\pi} = \frac{1}{8\pi Z}. \quad (5.3)$$

The existence of an isolated pole in the two-point function ensures, via the Källén–Lehmann representation, that the spectrum contains a single-particle state of mass zero. In the low-energy limit, cluster decomposition implies that connected higher-point correlation functions are dominated by exchanges of this single-particle state. The four-point function, in particular, factorises into a product of on-shell amplitudes and propagators at the pole. This factorisation follows from the pole structure, cluster decomposition, and the LSZ reduction formula, standard results of quantum field theory. The factorisation gives the pole structure required for Weinberg’s low-energy theorem, whose infrared premises are a massless pole, Lorentz invariance, and unitarity of the S -matrix.

5.2 Weinberg’s Low-Energy Theorem and Einstein Gravity

With the massless spin-2 propagator in the transverse-traceless sector established from the spectral pole in §5.1, we now turn to the question of how this pole translates into observable gravitational phenomena. In standard perturbative quantum field theory, one would proceed from the propagator to asymptotic states via the LSZ reduction formalism. For a universally coupled massless mode, however, the LSZ formalism is not the appropriate framework. Any attempt to isolate such a mode as a free asymptotic particle would require switching off its universal coupling, which is precisely what defines the mode and gives rise to its long-range effects. The observational content of a universally coupled spin-2 excitation is fully encoded in its static potential and its radiation, both of which are directly determined by the pole term of the two-point function and are accessible without constructing asymptotic free states. It reflects the distinctive nature of such excitations, which cannot be isolated without spoiling the very long-range correlations that characterize them.

The application of Weinberg’s low-energy theorem to the present emergent field does not require LSZ reduction. The theorem concerns the analytic structure of the S -matrix for matter–matter scattering—specifically, the long-range spin-2 component of such amplitudes. Its premises are the existence of a massless pole in the intermediate channel, Lorentz invariance, and unitarity of the S -matrix. Its conclusion—universal coupling with an Einstein–Hilbert low-energy effective action—is independent of whether the massless pole originates from an elementary field, a composite bound state, or an emergent collective mode of the type considered here.

Two observations place the use of the Weinberg theorem on the standard low-energy footing within the coarse-graining framework developed in this paper. First, at the infrared

fixed point $\sigma = \sigma_c$, the coarse-graining scale becomes spatially constant, $\sigma(x) \rightarrow \sigma_c$, and the window function reduces to $W_{\sigma_c}(p) = \exp(-p^2\sigma_c^2/2)$, which tends to unity for all momenta $p \ll \sigma_c^{-1}$. In this regime, the coarse-grained spectral density coincides with the full spectral density of the underlying quantum field theory in the infrared limit. Second, the loss of unitarity associated with integrating out fast modes is confined to scales above the cutoff σ_c^{-1} ; it does not affect the on-shell pole structure at $p^2 = 0$, and hence leaves the residue Z and the resulting long-range force unchanged. The low-energy S -matrix, restricted to external matter states with momentum $p \ll \sigma_c^{-1}$, therefore admits a unitary, Lorentz-invariant description to which the Weinberg theorem directly applies. This justification follows the standard effective field theory logic articulated in the seminal works of Weinberg on S -matrix constraints for massless particles [28] and of Donoghue on general relativity as an effective field theory [40]: low-energy observables—static potentials and radiation—are determined entirely by the on-shell poles and residues of the two-point function, and are insensitive to the ultraviolet details of the theory or to whether the massless mode can be prepared as an asymptotic free state.

The logical chain from the spectral pole to observable gravity can now be traced explicitly. The isolated massless pole in $\rho^{(2)}(\mu^2)$ yields, through the Källén–Lehmann representation, a massless spin-2 propagator in the transverse-traceless sector of the two-point function $\langle \Phi_{\mu\nu} \Phi_{\rho\sigma} \rangle$. From this propagator, the static limit reproduces the Newtonian $1/r$ potential, while the radiation limit yields the quadrupole formula for gravitational radiation. Finally, the Weinberg low-energy theorem, applied to the S -matrix of matter fields, fixes the universal coupling of this massless mode and uniquely determines the low-energy effective action to be of Einstein–Hilbert form. This sequence closes the connection between the spectral criterion and observable gravitational phenomena, without requiring the emergent field $\Phi_{\mu\nu}$ to possess asymptotic states in the LSZ sense. The observational consequences of the emergent spin-2 mode are fully captured by the long-range forces, gravitational radiation, and cosmological dynamics encoded in its two-point function and its universal coupling to matter, as derived in §3.2. Observational gravity is encoded in the two-point function, the static Newtonian potential and the quadrupole radiation formula, without requiring direct graviton detection.

The coupling of $h_{\mu\nu}$ to matter is the universal linear coupling derived in §3.2. After transverse-traceless projection, this takes the form $S_{\text{int}} = \kappa \int d^4x h_{\mu\nu}^{\text{TT}} T^{\mu\nu}$, with all matter fields coupling through the same energy–momentum tensor and the strength fixed by κ .

Weinberg’s theorem uniquely determines the nonlinear self-consistent completion. Starting from the three-point amplitude, unitarity and Lorentz invariance recursively fix all higher-point vertices, and the resulting S -matrix is equivalent at low energies to that of general relativity [28, 35]. The effective action that generates this S -matrix is the Einstein–Hilbert action. The effective metric $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ is the flat background metric modulated by the macroscopic collective mode of vacuum stress fluctuations. The field $h_{\mu\nu}$ itself carries energy and momentum and must therefore contribute its own energy–momentum tensor $t_{\mu\nu}[h]$. Self-consistency requires [1, 28],

$$\mathcal{E}_{\mu\nu}^{\rho\sigma} h_{\rho\sigma} = \kappa (T_{\mu\nu}^{\text{matter}} + t_{\mu\nu}[h]), \quad (5.4)$$

which is precisely the linearized Einstein equation with the gravitational field's own energy-momentum tensor included as a source. Expanding the resulting Einstein-Hilbert action to first order yields the linearized Einstein equation; in the static weak-field limit, its trace reduces to the Poisson equation $\nabla^2 h_{00} = -8\pi G \rho_m$, consistent with Newtonian gravity.

5.3 Newton's Constant and the Cosmological Constant

Once the pole exists, the effective Newton constant and the infrared curvature scale are encoded in the same spectral data and coarse-graining scales that govern the flow. No independent fundamental spin-2 coupling is introduced.

Newton's constant. When the spin-2 spectral density develops an isolated pole at zero momentum, the pole residue Z fixes the coupling strength of the emergent massless spin-2 field through $G = 1/(8\pi Z)$, as shown in Appendix C. The residue is formed at the emergence scale σ_G , where all Standard Model degrees of freedom are active. Dimensional analysis and the spectral representation give $Z \sim N_{\text{eff}}/M_G^2$, where $M_G = 1/\sigma_G$ is the emergence energy scale and N_{eff} is the effective number of polarization-weighted degrees of freedom. Counting the gauge bosons, fermions, and the Higgs doublet of the Standard Model yields $N_{\text{eff}} \approx 100$. (For each active massless gauge boson, the two transverse polarizations contribute with weight 2. For each Weyl fermion, the two helicity states contribute with weight 2. For the Higgs doublet, the four real scalar components contribute with weight 4. The precise value depends on the number of active degrees of freedom at the emergence scale and does not affect the order-of-magnitude conclusion.) Hence

$$G = \frac{1}{8\pi Z} \sim \frac{M_G^2}{8\pi N_{\text{eff}}}. \quad (5.5)$$

The dimensionless combination $8\pi G/M_G^2$ is of order $1/N_{\text{eff}} \sim 10^{-2}$, consistent with the weakness of the long-range coupling relative to the emergence scale. The emergence scale M_G is not fixed a priori in this paper. The observed value of G constrains it, while a first-principles value would require a nonperturbative calculation of the pole residue Z . The estimate above should therefore be read as a spectral scaling relation, not as a completed numerical prediction.

Cosmological constant. At the infrared fixed point $\sigma_c = c/H_0$, the effective temperature has dropped to

$$k_B T_{\text{eff}}(\sigma_c) = \frac{\hbar c}{2\pi\sigma_c} = \frac{\hbar H_0}{2\pi}. \quad (5.6)$$

All massive degrees of freedom are frozen out. The infrared endpoint supplies the natural curvature scale of the effective description. Dimensional analysis of the effective potential at the fixed point yields

$$\Lambda \sim \frac{1}{\sigma_c^2} = \left(\frac{H_0}{c}\right)^2 \sim 10^{-52} \text{ m}^{-2}, \quad (5.7)$$

corresponding to an energy density $\rho_\Lambda \sim M_P^2 \Lambda \sim 10^{-47} \text{ GeV}^4$, where the factor M_P^2 arises from dimensional analysis. The quantity Λ has dimensions of $(\text{length})^{-2}$, and the Planck mass is the natural ultraviolet scale of the effective field theory. This value is consistent

with the observed dark energy density. A positive curvature scale corresponds to a de Sitter infrared attractor, with the effective potential minimized at σ_c .

The commutator spectral criterion is insensitive to additive vacuum-energy constants: such constants do not appear in the propagating spin-2 spectral density. The hierarchy between the Planck-scale vacuum density and the observed dark-energy scale is therefore not obtained by summing ultraviolet zero-point energies and then cancelling them. In this framework it is encoded in the separation between the ultraviolet and infrared endpoints of the coarse-graining flow. A complete semiclassical treatment of the vacuum expectation value after the emergent metric has formed is a separate effective-gravity question.

Hawking temperature. The effective temperature $T_{\text{eff}} = \hbar c / (2\pi\sigma)$ also provides a consistency check with known gravitational thermodynamics. The horizon of a Schwarzschild black hole acts as an infrared boundary for vacuum fluctuations, much as the Hubble horizon does at cosmological scales. Taking $\sigma \sim 2r_S = 4GM/c^2$ in the effective temperature formula (3.16) yields

$$k_B T_H = \frac{\hbar c}{4\pi r_S} = \frac{\hbar c^3}{8\pi GM}, \quad (5.8)$$

the standard Hawking temperature. In this framework, this relation is read as the coarse-graining temperature evaluated at the black-hole horizon scale.

5.4 Falsifiability of the Criterion

The criterion has clear failure modes. If a nonperturbative calculation of the stress-tensor spectral density finds no isolated pole at $p^2 = 0$, the coarse-grained tensor field does not mediate long-range Einstein gravity. If the spin-2 channel contains only a continuum or a massive resonance, the resulting force is not the observed long-range interaction. If the pole appears with negative residue, unitarity is lost. If the pole exists but the low-energy coupling is not universal, the assumptions needed for Weinberg’s theorem are not met. In each case the spectral construction still defines a coarse-grained stress-fluctuation field, but it does not yield emergent gravity.

6 Summary

The criterion proposed in this paper is whether the spin-2 spectral density of the energy-momentum tensor commutator develops an isolated zero-momentum pole. This is the proposed signature of emergent gravity from quantum field fluctuations.

The irreversibility of coarse-graining drives the construction. Discarding fast modes produces energy transfer and entropy production. Their ratio defines an effective temperature; in the free massless zero-mode limit this gives $k_B T_{\text{eff}} = \hbar c / (2\pi\sigma)$. The same trace density that yields temperature also redistributes spectral weight among the surviving slow modes. The effective temperature encodes the statistical effect of the discarded modes on the scalar channel. The spectral pole—should it form—encodes their effect on the spin-2 channel. Two macroscopic parameters, one microscopic substrate.

The central criterion is the selection of the δ -function pole. Within the spectral-flow class considered here, smooth spectral functions are obstructed as fixed points. This

negative result makes the pole question sharp. If the pole forms, Weinberg’s low-energy theorem fixes the effective action to the Einstein–Hilbert form. Newton’s constant follows from the pole residue, $G = 1/(8\pi Z)$. The infrared fixed-point scale gives $\Lambda \sim (H_0/c)^2$. These relations give the observed orders of magnitude when the residue and the endpoint scales take their natural values. The effective temperature, evaluated at the Schwarzschild horizon scale, returns the Hawking temperature, serving as a consistency check rather than an input. Lattice gauge theory and nonperturbative functional renormalization-group methods provide direct tests of the pole condition.

The equivalence principle follows, in the pole phase, from the Lorentz structure of the Standard Model and the uniqueness of the energy–momentum tensor as the spin-2 current. The Einstein equivalence principle follows from the uniqueness of the effective geometry, and the strong equivalence principle from the nonlinear completion fixed by Weinberg’s theorem. Temperature and gravity are not merely analogous. The Introduction drew the parallel: gravity might stand to the energy–momentum tensor as temperature stands to molecular motion. The intervening chapters made it precise as a spectral criterion. The logic that leads from coarse-graining to temperature is the same logic that leads from coarse-graining to the spin-2 pole question. They are two channels of a single trace density. The commutator, by restricting to the fluctuating sector, encodes this at the kinematic level. Gravity is not pre-installed; it is selected by the infrared pole structure.

Should the pole form, the resulting massless spin-2 excitation is not a fundamental particle. It is the quantum of the emergent order parameter $h_{\mu\nu}$. The non-renormalisability encountered when quantising gravity as a fundamental field is therefore not the starting point of the construction; what is quantised are collective modes of energy–momentum tensor fluctuations.

Recent computations in Lorentzian quantum gravity have found a massless one-graviton peak and a multi-graviton continuum [39, 41, 42], a spectral structure that shares the same qualitative structure as the one conjectured here, despite the different spacetime background. This suggests that such a structure is realizable in a nonperturbative quantum field theory.

Several directions lie beyond the criterion established here. Reconstruction of the full nonlinear effective action from complete spectral data is a broader inverse problem. The coarse-graining analysis is Euclidean, while the Lorentzian causal structure belongs to the corresponding real-time formulation. The large- N model of Appendix B illustrates the mechanism in a controlled setting, but spin-2 is not spin-1. In the full Standard Model, the stronger transversality condition supplies the specific spin-2 test of the pole mechanism. These directions define the programme beyond the present criterion. This paper fixes the first step. If the pole exists, the parallel between temperature and gravity is complete. If it does not, the parallel fails.

The emergence question is therefore reduced to the spectral pole criterion.

A Fast-Mode Integration and the Langevin Equation

This appendix demonstrates how the Langevin equation arises naturally within the coarse-graining framework. It is not a rigorous derivation from the Wetterich equation. The complete treatment of composite-operator FRG flows, real-time continuation, and non-Markovian corrections is deferred to dedicated technical work. The following derivation is qualitatively valid in the limit of a sufficiently steep regulator.

A.1 Fast-slow mode decomposition

The energy-momentum tensor operator $\hat{T}_{\mu\nu}$ is separated by momentum into slow modes ($p < k$) and fast modes ($p > k$), where k is the coarse-graining scale. Denote the slow-mode component by σ (representing a typical component of the order-parameter field $h_{\mu\nu}$), and the fast-mode component as the environmental degrees of freedom. The rigorous implementation of this decomposition depends on the explicit form of the regulator R_k (see Section 2 of the main text).

A.2 Path-integral formulation and fast-mode integration

The Euclidean generating functional of the system is

$$Z = \int \mathcal{D}\sigma \int \mathcal{D}[\text{fast}] \exp(-S[\sigma] - S_{\text{env}}[\text{fast}] - S_{\text{int}}[\sigma, \text{fast}]), \quad (\text{A.1})$$

where $S[\sigma]$ and $S_{\text{env}}[\text{fast}]$ are the free actions of the slow and fast modes respectively, and S_{int} is the coupling term. Expanding the fast modes in normal coordinates $\{q_n\}$, where n labels the complete set of fast-mode eigenstates, the interaction action contains, at lowest order, a linear coupling term,

$$S_{\text{int}}[\sigma, q] = \sum_n C_n \cdot \sigma \cdot q_n + \dots, \quad (\text{A.2})$$

where C_n is given by the matrix element of the energy-momentum tensor between slow and fast modes. Near the critical point, σ is small and the linear term is a reasonable starting point. Higher-order coupling terms are suppressed in the vicinity of the critical point, but may become important farther away from it.

A.3 Fast-mode Green's function and effective action

Integrating out the fast modes as a Gaussian integral yields the effective action for the slow modes,

$$S_{\text{eff}}[\sigma] = S[\sigma] + \frac{1}{2} \text{Tr} \ln(G_{\text{fast}}^{-1} + V[\sigma]), \quad (\text{A.3})$$

where G_{fast} is the bare Green's function of the fast modes and $V[\sigma]$ is the coupling potential. Expanding to second order in the slow-variation approximation gives an effective action containing a nonlocal kernel,

$$S_{\text{eff}}^{(2)}[\sigma] = \frac{1}{2} \int d\tau \int d\tau' \sigma(\tau) \Sigma(\tau - \tau') \sigma(\tau'), \quad (\text{A.4})$$

where $\Sigma(\tau - \tau')$ is the fast-mode polarization tensor represented in the time domain. The analytic properties of its Fourier transform $\Sigma(\omega)$ determine the structure of the dissipation kernel.

A.4 From effective action to Langevin equation

To convert the nonlocal action into a local equation, an auxiliary stochastic force $\xi(\tau)$ is introduced via a Hubbard–Stratonovich transformation,

$$e^{-\frac{1}{2} \int \sigma \Sigma \sigma} \propto \int \mathcal{D}\xi \exp\left(-\frac{1}{2} \int \xi \Sigma^{-1} \xi + i \int \xi \sigma\right). \quad (\text{A.5})$$

Positivity of the kernel Σ in the Euclidean formulation is guaranteed by the positivity of the fast-mode spectral function (required by unitarity). After continuation to real time, the saddle-point condition yields the Langevin equation

$$\partial_t^2 \sigma + \int_{-\infty}^t dt' \Gamma(t - t') \partial_{t'} \sigma(t') + \frac{\delta V_{\text{eff}}[\sigma]}{\delta \sigma} = \xi(t), \quad (\text{A.6})$$

where the dissipation kernel $\Gamma(t - t')$ is determined by the imaginary part of the retarded self-energy,

$$\Gamma(\omega) = \Im \Sigma_{\text{ret}}(\omega). \quad (\text{A.7})$$

The noise correlator is

$$\langle \xi(t) \xi(t') \rangle = \nu(t - t'), \quad (\text{A.8})$$

where $\nu(\omega)$ is given by the real part of the fast-mode spectral function. For a quantum environment, $\nu(\omega)$ and $\Gamma(\omega)$ satisfy the fluctuation–dissipation theorem,

$$\nu(\omega) = \hbar \coth\left(\frac{\hbar \omega}{2k_B T}\right) \cdot \Gamma(\omega). \quad (\text{A.9})$$

A.5 Overdamped limit and effective temperature

Near the coarse-graining critical point, the relaxation time of the slow mode is substantially longer than its oscillation period. In the overdamped limit the inertial term can be neglected and the dissipation kernel can be approximated by its zero-frequency limit $\Gamma = \Gamma(\omega \rightarrow 0)$. The equation reduces to

$$\Gamma \cdot \partial_t \sigma + \frac{\delta V_{\text{eff}}[\sigma]}{\delta \sigma} = \xi(t). \quad (\text{A.10})$$

Fast modes are continuously integrated out during coarse-graining, and their effect is equivalent to a heat bath at an effective temperature T_{eff} . Under the condition $k_B T_{\text{eff}} \gg \hbar \omega$, the quantum fluctuation–dissipation theorem reduces to its classical form,

$$\nu(\omega) = 2k_B T_{\text{eff}} \cdot \Gamma(\omega). \quad (\text{A.11})$$

The applicability of this relation near the emergence scale relies on scale separation. Slow-mode frequencies are far below the effective temperature, a condition that is satisfied near the critical point. At lower infrared scales, when the slow-mode frequency becomes comparable to the temperature, the quantum fluctuation–dissipation relation must be restored.

A.6 Limitations

The derivation above is simplified at several levels, and the regime of validity of each approximation requires further investigation.

The strict separation of fast and slow modes is only approximate for a smooth regulator. The derivation is qualitatively valid. Quantitative details depend on the specific choice of regulator scheme.

The truncation to linear coupling may require nonlinear corrections when σ is large, far from the critical point. Near the critical point these corrections are negligible.

The Markovian approximation, $\Gamma(\omega) \rightarrow \Gamma(0)$, breaks down exactly at the critical point. The present framework addresses the flow near the critical point, not the behavior exactly on it. When the system sits precisely at criticality, a more involved non-Markovian description is needed.

The continuation from Euclidean to real time involves the standard analytic continuation problem of quantum field theory, which is not elaborated here.

The exponential regulator used in this paper (see Section 2 of the main text) provides a concrete implementation of the fast-slow mode decomposition. Within this scheme, the qualitative conclusions of this appendix, namely the existence of dissipative and noise terms, the emergence of an effective temperature, and the applicability of the fluctuation-dissipation relation, are robust. Quantitative details depend on the regulator scheme, but do not affect the central criterion of the main text.

These caveats do not undermine the purpose of this appendix. It shows that the Langevin equation follows from the framework of fast-mode integration.

B Pole Formation in the Large- N Limit

This appendix provides a well-defined, exactly solvable example of the central mechanism, rather than a rigorous proof that the pole appears in the full theory.

B.1 Model definition

Consider a scalar field theory with $O(N)$ symmetry in four-dimensional Euclidean space-time:

$$S = \int d^4x \left[\frac{1}{2} \partial_\mu \phi^a \partial_\mu \phi^a + \frac{1}{2} m^2 \phi^a \phi^a + \frac{\lambda}{4!N} (\phi^a \phi^a)^2 \right], \quad (\text{B.1})$$

where $a = 1, \dots, N$. In the large- N limit ($N \rightarrow \infty$, λ fixed), this theory can be solved exactly via the Schwinger-Dyson equations.

We focus on the conserved $O(N)$ Noether current,

$$J_\mu^{ab} = \phi^a \partial_\mu \phi^b - \phi^b \partial_\mu \phi^a. \quad (\text{B.2})$$

The two-point function of this conserved current bears structural similarity to that of the energy-momentum tensor. Both are conserved currents (of spin-1 and spin-2 respectively) whose spectral functions are subject to superconvergence sum rules. We use the Noether current here instead of the energy-momentum tensor itself because the former is exactly

tractable in the large- N limit. We expect that the two systems share the same qualitative mechanism of pole formation. The extent to which this expectation carries over to the full theory is a quantitative question discussed in B.5.

B.2 Schwinger–Dyson equation and spectral representation

In the large- N limit, the two-point function of the conserved current satisfies a closed Schwinger–Dyson equation. The spectral representation of the full propagator is

$$G(p) = \int_0^\infty \frac{d\mu^2}{p^2 + \mu^2} \rho(\mu^2), \quad (\text{B.3})$$

with the spectral function satisfying positivity: $\rho(\mu^2) \geq 0$. In the free-field limit, the spectral function is

$$\rho_0(\mu^2) = \delta(\mu^2 - m^2). \quad (\text{B.4})$$

In the interacting theory, the spectral function satisfies the integral equation

$$\rho(s) = \rho_0(s) + \frac{1}{N} \int_0^\infty ds' \mathcal{K}(s, s') \rho(s'), \quad (\text{B.5})$$

where $s = \mu^2$. The integral kernel $\mathcal{K}(s, s')$ is given by the one-loop diagram. In the large- N limit, higher-loop contributions are suppressed by powers of $1/N$; the one-loop result is exact. The explicit structure of the kernel contains a logarithmic divergence as

(s

to 0

). This is the mathematical origin of the nontrivial low-energy behavior of the conserved current.

B.3 Coarse-graining and the FRG flow

Introducing a coarse-graining scale k and a regulator R_k , the spectral function acquires a dependence on k : $\rho(s) \rightarrow \rho_k(s)$. The FRG flow equation is

$$k \frac{\partial}{\partial k} \rho_k(s) = \int_0^\infty ds' \mathcal{K}_k(s, s') \rho_k(s'), \quad (\text{B.6})$$

where the flow kernel $\mathcal{K}_k(s, s')$ is the original integral kernel deformed by coarse-graining, containing regulator-derivative factors $f_k(s)$:

$$\mathcal{K}_k(s, s') = \mathcal{K}(s, s') \cdot f_k(s) f_k(s'). \quad (\text{B.7})$$

For the exponential regulator $R_k(z) = z/(e^{z/k^2} - 1)$,

$$f_k(s) = \frac{s}{k^2} \cdot \frac{e^{s/k^2}}{(e^{s/k^2} - 1)^2}. \quad (\text{B.8})$$

This factor decays exponentially for $s \gg k^2$ and vanishes as $s \rightarrow 0$. The specific choice of regulator affects the quantitative behavior of the flow kernel. The logarithmic low-energy structure of the conserved-current kernel is the feature used below.

B.4 Verification of pole formation

We demonstrate here that in the large- N limit, a simplified, exactly tractable model, the spectral function spontaneously develops a δ -function pole under the coarse-graining flow. This is not a rigorous proof of pole formation in the full theory, but a well-defined illustration of the central mechanism.

Decompose the spectral function into a pole part and a continuum part:

$$\rho_k(s) = A_k \delta(s) + \tilde{\rho}_k(s), \quad (\text{B.9})$$

where $\tilde{\rho}_k(s)$ is the continuum. Substituting this decomposition into the FRG flow equation, we examine the evolution of the pole term and the continuum separately.

For the pole weight A_k , we insert the decomposition into the flow equation and take the limit $s \rightarrow 0$. The flow kernel $\mathcal{K}_k(s, s')$ behaves as $1/s$ as $s \rightarrow 0$. This singular behavior originates from the kinematic constraints on the conserved-current propagator at zero momentum. The continuum part $\tilde{\rho}_k(s')$ transfers spectral weight to $s = 0$ through the kernel, contributing an inhomogeneous term. Simultaneously, the conserved current obeys a sum rule that enforces the conservation of total spectral weight:

$$\int_0^\infty ds \rho_k(s) = \text{const} \quad (\text{independent of } k). \quad (\text{B.10})$$

This sum rule is a direct consequence of the conservation law in the spectral representation. It takes the form $\int ds \rho_k(s) = \text{const}$ for a spin-1 conserved current, rather than the weighted form $\int ds s \cdot \rho_k(s) = \text{const}$ appropriate to the spin-2 energy-momentum tensor.

Combining these two effects yields the evolution equation for A_k :

$$k \frac{\partial A_k}{\partial k} = \gamma_k A_k + \beta_k, \quad (\text{B.11})$$

where γ_k is given by the residue of the integral kernel at the pole, and $\beta_k > 0$ describes the feedback from the continuum onto the pole. The positivity of β_k follows from the positivity of the spectral function ($\tilde{\rho}_k(s) \geq 0$) and the positive singular behavior of the flow kernel as $s \rightarrow 0$.

The structure of the equation carries a clear physical meaning: the term $\gamma_k A_k$ describes the scaling of the existing pole weight under the flow, while β_k describes the continuous transfer of weight from the continuum to the pole. Even if $A_\Lambda = 0$ initially (no pole at the ultraviolet cutoff), $\beta_k > 0$ generates a non-zero A_k during the coarse-graining flow in this model. The appearance of the pole is a consequence of the structure of the flow equation and the constraints imposed by the conservation law, not a fine-tuning of initial conditions.

In the large- N limit, β_k remains positive for all finite k . Starting from any smooth initial spectral function, the pole weight A_k is necessarily driven to a non-zero value in the infrared limit of the coarse-graining flow. This is the central mathematical fact of pole formation in the large- N model.

B.5 Relation to the full theory

Two differences exist between the present model and the energy-momentum tensor system studied in the main text.

- (1) The Noether current is a spin-1 operator (antisymmetric in the internal indices), whereas the energy-momentum tensor is a spin-2 operator. The transversality condition for spin-2 is stronger than the conservation condition for spin-1. It eliminates not only the longitudinal component but also the trace component. This means that the spectral function of the energy-momentum tensor is subject to more kinematic constraints. Our preliminary assessment is that stronger constraints make it more difficult for a smooth function to satisfy all conditions, thereby making the appearance of a pole more likely. The correctness of this assessment remains to be verified in the full theory.
- (2) The large- N limit suppresses all loop contributions beyond one-loop order. In the full theory at finite N , higher-order loop corrections may quantitatively alter the critical scale at which the pole forms, or prevent the threshold from being reached. The qualitative constraints imposed by conservation and unitarity on the spectral function—positivity, sum rules, and low-energy kinematic singularities—hold at any order. The large- N model therefore demonstrates a controlled mechanism, not a general proof for the full spin-2 theory.

B.6 Falsifiability

This model provides a clear path toward testing the central criterion of the main text. In the large- N model, pole formation follows from the structure of the flow equation. If, in the full theory, lattice simulations or nonperturbative FRG numerical calculations find that the spin-2 spectral density of the energy-momentum tensor remains smooth at low energies without developing an isolated pole, the proposed gravity mechanism is absent. At minimum, the large- N model demonstrates that the core mechanism operates in a well-defined setting, providing motivation for searching for the same mechanism in more complex situations.

B.7 Remarks on the Mechanism in QCD

The pole-formation mechanism studied above in the large- N $O(N)$ model has a well-known analogue in QCD. This observation is an analogy. It is not independent evidence for emergent gravity. It serves to illustrate that the type of integral-equation-driven pole formation examined in Appendix B is not an artifact of our particular toy model. It is a recurring feature of non-perturbative quantum field theory.

In the large- N_c limit of QCD, meson bound states appear as poles in the Green's functions. Their spectrum is determined by an integral equation—the 't Hooft equation [43]—which follows from the Bethe-Salpeter equation for the quark-antiquark amplitude [44, 45]. The equation takes the form of a self-consistent integral equation for the meson spectral density. The kernel accumulates spectral weight until a discrete pole emerges. This is structurally the same feedback mechanism demonstrated in (B.11), though in a different spacetime dimension.

Consider the spin-2 tensor meson $f_2(1270)$. Its spectral function is conventionally parametrized as

$$\rho_{\text{QCD}}^{(2)}(s) = F_R \delta(s - m_{f_2}^2) + \rho_{\text{cont}}(s), \quad (\text{B.12})$$

with $m_{f_2} = 1.2755 \pm 0.0011$ GeV [46]. The continuum $\rho_{\text{cont}}(s)$ is described by QCD sum rules [8, 47], where the gluon condensate provides the leading non-perturbative contribution. The existence of this resonance is well established experimentally.

The QCD mechanism is not identical to the one in our toy model in all details. The 't Hooft equation operates in $1+1$ dimensions, whereas the large- N model in Appendix B is formulated in four-dimensional Euclidean space. What the two share is a common qualitative mechanism: under coarse-graining or in the large- N limit, the spectral density satisfies an integral equation whose kernel drives a self-consistent accumulation of spectral weight until a δ -function pole emerges. The QCD case shows that feedback-driven pole formation is a familiar nonperturbative phenomenon in quantum field theory. It does not by itself establish the spin-2 gravitational pole proposed in the main text.

The analogy has clear limitations. The spectral pole in QCD arises from the specific dynamics of gauge interactions—gluon exchange between quarks. This is kinematically distinct from the coarse-graining of energy-momentum tensor vacuum fluctuations in the main text. The $f_2(1270)$ is a quark-antiquark bound state, not an emergent graviton. What the analogy establishes is the mathematical structure of feedback-driven pole formation as a genuine feature of QFT. It does not establish that QCD already exhibits emergent gravity. The distinction between gauge-driven bound-state poles and coarse-graining-driven spectral poles should be kept clear.

What distinguishes the strong-interaction case from the gravitational case is the **location** of the pole. In QCD, gauge dynamics fixes the pole at $s = m_R^2 > 0$. For the energy-momentum tensor in the main text, the Ward identity and the superconvergence sum rule (4.5) select $s = 0$ as the location of an isolated long-range pole. The framework therefore accommodates massive composite spin-2 resonances in QCD and a possible massless emergent spin-2 mode under a unified spectral language. The hierarchy between the QCD scale and the gravitational scale reflects the different symmetry structures that determine where the feedback-driven pole is kinematically allowed to form.

C From Spectral Pole to Massless Propagator

This appendix provides a detailed derivation of the connection between the spectral pole and the massless propagator. If the coarse-grained spectral density $\rho^{(2)}(\mu^2)$ contains an isolated pole at zero momentum, then upon inserting it into the spectral representation, the transverse-traceless sector of the correlation function acquires a pole term Z/p^2 , corresponding to a massless spin-2 propagator.

Suppose the coarse-grained spectral density possesses an isolated pole at zero momentum:

$$\rho^{(2)}(\mu^2) = Z\delta(\mu^2) + \tilde{\rho}^{(2)}(\mu^2), \quad (\text{C.1})$$

where $Z > 0$ is the pole residue and $\tilde{\rho}^{(2)}(\mu^2)$ is the smooth continuum part. Insert (C.1) into the spin-2 sector of the spectral representation (2.10):

$$\langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma = \int_0^\infty d\mu^2 \rho^{\text{cg}}(\mu^2; \sigma) P_{\mu\nu\alpha\beta}^{\text{TT}} \Delta(x-y; \mu^2). \quad (\text{C.2})$$

After inserting the pole ansatz, the continuum part contributes a regular term, and the pole contribution is

$$\langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma \supset Z \int_0^\infty d\mu^2 \delta(\mu^2) P_{\mu\nu\alpha\beta}^{\text{TT}} \Delta(x-y; \mu^2). \quad (\text{C.3})$$

Using the property $\int_0^\infty d\mu^2 \delta(\mu^2) f(\mu^2) = f(0)$, the integral need only be evaluated at $\mu^2 = 0$. The projector $P_{\mu\nu\alpha\beta}^{\text{TT}}$ is independent of μ^2 and can be pulled out of the integral, giving

$$\langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma \supset Z P_{\mu\nu\alpha\beta}^{\text{TT}} \Delta(x-y; 0). \quad (\text{C.4})$$

The massless scalar propagator in coordinate space takes the form

$$\Delta(x-y; 0) = \frac{1}{4\pi^2 |x-y|^2}. \quad (\text{C.5})$$

Substituting (C.5) into (C.4) gives the coordinate-space correlation function

$$\langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma \supset \frac{Z}{4\pi^2 |x-y|^2} P_{\mu\nu\alpha\beta}^{\text{TT}}. \quad (\text{C.6})$$

To convert to momentum space, use the Fourier transform of the four-dimensional massless scalar propagator:

$$\frac{1}{4\pi^2 |x-y|^2} = \int \frac{d^4 p}{(2\pi)^4} \frac{e^{ip \cdot (x-y)}}{p^2}. \quad (\text{C.7})$$

Inserting (C.7) into (C.6) yields

$$\langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma \supset Z \int \frac{d^4 p}{(2\pi)^4} \frac{e^{ip \cdot (x-y)}}{p^2} P_{\mu\nu\alpha\beta}^{\text{TT}}(p). \quad (\text{C.8})$$

In momentum space the two-point function is defined as

$$\langle \Phi_{\mu\nu}^{\text{TT}}(p) \Phi_{\alpha\beta}^{\text{TT}}(-p) \rangle_\sigma = \int d^4(x-y) e^{-ip \cdot (x-y)} \langle \Phi_{\mu\nu}^{\text{TT}}(x) \Phi_{\alpha\beta}^{\text{TT}}(y) \rangle_\sigma. \quad (\text{C.9})$$

Substituting (C.8) into (C.9) and using $\int d^4 x e^{-i(p-q) \cdot x} = (2\pi)^4 \delta^{(4)}(p-q)$ gives

$$\langle \Phi_{\mu\nu}^{\text{TT}}(p) \Phi_{\alpha\beta}^{\text{TT}}(-p) \rangle_\sigma \supset \frac{Z}{p^2} P_{\mu\nu\alpha\beta}^{\text{TT}}(p). \quad (\text{C.10})$$

This is Eq. (5.1) of the main text. In coordinate space this pole term is equivalent to the Pauli–Fierz action [1]:

$$S_{\text{PF}} = \frac{1}{2\kappa^2} \int d^4 x h^{\mu\nu} E_{\mu\nu}^{\rho\sigma} h_{\rho\sigma}, \quad (\text{C.11})$$

where $h_{\mu\nu} \equiv \kappa \Phi_{\mu\nu}^{\text{TT}}$, $\kappa^2 = 1/Z$, and $E_{\mu\nu}^{\rho\sigma}$ is the linearized Einstein tensor operator. Newton's constant G is read from the pole residue:

$$G = \frac{\kappa^2}{8\pi} = \frac{1}{8\pi Z}. \quad (\text{C.12})$$

The existence of an isolated zero-momentum pole in the spectral density is equivalent to the macroscopic field $\Phi_{\mu\nu}^{\text{TT}}$ acquiring a massless propagator. If the spectral density contains no isolated pole at $p^2 = 0$, the pole contribution in (C.3) is absent. The transverse-traceless sector then has no isolated massless exchange term and does not produce a macroscopic Newtonian $1/r$ force.

D Evaluation of the Entropy and Energy Kernels

This appendix carries out the integrals that appear in the entropy and energy kernels of Sec. 3.3. The regulator is the exponential window

$$R_k(z) = \frac{z}{e^{z/k^2} - 1}, \quad z = p_0^2 + \lambda. \quad (\text{D.1})$$

The derivative with respect to RG time $t = \ln k$ is

$$\partial_t R_k(z) = \frac{2z^2}{k^2} \frac{e^{z/k^2}}{(e^{z/k^2} - 1)^2}. \quad (\text{D.2})$$

Both kernels are evaluated in the free-field, massless, zero-mode limit $\lambda = 0$, $m^2 = 0$, for which $z = p_0^2$. The two integrals to be computed are

$$\mathcal{I}_E = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k(p_0^2)}{(p_0^2 + R_k(p_0^2))^2}, \quad \mathcal{I}_S = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{\partial_t R_k(p_0^2)}{p_0^2 + R_k(p_0^2)}. \quad (\text{D.3})$$

D.1 Energy kernel

From (D.1) with $z = p_0^2$,

$$R_k(p_0^2) = \frac{p_0^2}{e^{p_0^2/k^2} - 1}, \quad \partial_t R_k(p_0^2) = \frac{2p_0^4}{k^2} \frac{e^{p_0^2/k^2}}{(e^{p_0^2/k^2} - 1)^2}. \quad (\text{D.4})$$

The denominator of the energy kernel is

$$p_0^2 + R_k(p_0^2) = p_0^2 + \frac{p_0^2}{e^{p_0^2/k^2} - 1} = p_0^2 \frac{e^{p_0^2/k^2}}{e^{p_0^2/k^2} - 1}. \quad (\text{D.5})$$

Squaring gives

$$(p_0^2 + R_k(p_0^2))^2 = p_0^4 \frac{e^{2p_0^2/k^2}}{(e^{p_0^2/k^2} - 1)^2}. \quad (\text{D.6})$$

The integrand of $\mathcal{I}_E$ is the ratio of (D.4) to (D.6):

$$\frac{\partial_t R_k}{(p_0^2 + R_k)^2} = \frac{2p_0^4}{k^2} \frac{e^{p_0^2/k^2}}{(e^{p_0^2/k^2} - 1)^2} \cdot \frac{(e^{p_0^2/k^2} - 1)^2}{p_0^4 e^{2p_0^2/k^2}} = \frac{2}{k^2} e^{-p_0^2/k^2}. \quad (\text{D.7})$$

The p_0^4 factors and the squared Bose factors cancel exactly. The integral is Gaussian:

$$\mathcal{I}_E = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{2}{k^2} e^{-p_0^2/k^2} = \frac{1}{\pi k^2} \int_0^{\infty} dp_0 e^{-p_0^2/k^2}. \quad (\text{D.8})$$

With $\int_0^{\infty} e^{-p_0^2/k^2} dp_0 = \sqrt{\pi} k/2$,

$$\mathcal{I}_E = \frac{1}{\sqrt{\pi} k}. \quad (\text{D.9})$$

D.2 Entropy kernel

The denominator of the entropy kernel is (D.5):

$$p_0^2 + R_k(p_0^2) = p_0^2 \frac{e^{p_0^2/k^2}}{e^{p_0^2/k^2} - 1}. \quad (\text{D.10})$$

The integrand of $\mathcal{I}_S$ is the ratio of (D.4) to (D.10):

$$\frac{\partial_t R_k}{p_0^2 + R_k} = \frac{2p_0^4}{k^2} \frac{e^{p_0^2/k^2}}{(e^{p_0^2/k^2} - 1)^2} \cdot \frac{e^{p_0^2/k^2} - 1}{p_0^2 e^{p_0^2/k^2}} = \frac{2p_0^2}{k^2} \frac{1}{e^{p_0^2/k^2} - 1}. \quad (\text{D.11})$$

One factor of p_0^2 , one factor of $e^{p_0^2/k^2}$, and one power of the Bose factor cancel. The integral is

$$\mathcal{I}_S = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} \frac{2p_0^2}{k^2} \frac{1}{e^{p_0^2/k^2} - 1} = \frac{2}{\pi k^2} \int_0^{\infty} dp_0 \frac{p_0^2}{e^{p_0^2/k^2} - 1}. \quad (\text{D.12})$$

Set $x = p_0/k$, so $p_0 = kx$, $dp_0 = k dx$, $p_0^2 = k^2 x^2$. Then

$$\mathcal{I}_S = \frac{2}{\pi k^2} \int_0^{\infty} k dx \frac{k^2 x^2}{e^{x^2} - 1} = \frac{2k}{\pi} \int_0^{\infty} \frac{x^2}{e^{x^2} - 1} dx. \quad (\text{D.13})$$

To evaluate the remaining integral, set $u = x^2$, so $x = \sqrt{u}$, $dx = du/(2\sqrt{u})$:

$$\int_0^{\infty} \frac{x^2}{e^{x^2} - 1} dx = \int_0^{\infty} \frac{u}{e^u - 1} \frac{du}{2\sqrt{u}} = \frac{1}{2} \int_0^{\infty} \frac{\sqrt{u}}{e^u - 1} du. \quad (\text{D.14})$$

The integral $\int_0^{\infty} \sqrt{u}/(e^u - 1) du$ is a standard Bose integral. It evaluates to $\Gamma(3/2) \zeta(3/2)$, with $\Gamma(3/2) = \sqrt{\pi}/2$ and $\zeta(3/2) \approx 2.612$. Hence

$$\int_0^{\infty} \frac{x^2}{e^{x^2} - 1} dx = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} \zeta(3/2) = \frac{\sqrt{\pi}}{4} \zeta(3/2). \quad (\text{D.15})$$

Substituting back into (D.13),

$$\mathcal{I}_S = \frac{2k}{\pi} \cdot \frac{\sqrt{\pi}}{4} \zeta(3/2) = \frac{k}{2\sqrt{\pi}} \zeta(3/2). \quad (\text{D.16})$$

So

$$\mathcal{I}_S = \frac{k}{2\sqrt{\pi}} \zeta(3/2). \quad (\text{D.17})$$

D.3 Ratio of the two kernels

From (D.9) and (D.17),

$$\frac{\mathcal{I}_S}{\mathcal{I}_E} = \frac{k}{2\sqrt{\pi}} \zeta(3/2) \cdot \frac{\sqrt{\pi} k}{1} = \frac{k^2}{2} \zeta(3/2). \quad (\text{D.18})$$

Numerically, $\zeta(3/2) \approx 2.612$, so the ratio is approximately $1.306 k^2$.

D.4 Comparison with the Litim regulator

For the Litim sharp cutoff $R_k(z) = (k^2 - z) \theta(k^2 - z)$, the analogous calculation gives

$$\mathcal{I}_E^{\text{Litim}} = \frac{1}{2\pi k}, \quad \mathcal{I}_S^{\text{Litim}} = \frac{k}{2\pi}, \quad \frac{\mathcal{I}_S^{\text{Litim}}}{\mathcal{I}_E^{\text{Litim}}} = k^2. \quad (\text{D.19})$$

The ratio is exactly k^2 with no zeta-function factor. The difference between (D.18) and (D.19) illustrates the regulator dependence of the numerical coefficient. The scaling $\mathcal{I}_S/\mathcal{I}_E \propto k^2$ is universal.

E Feynman Diagram Classification of the Injection Term

E.1 Definition of the Injection Term

The injection term $\Delta_\sigma(\mu^2)$ is introduced in the main text in § 2.3. It is defined as the contribution to the spin-2 spectral density from the momentum shell $[1/\sigma, 1/(\sigma + d\sigma)]$ after the fast modes in that shell are integrated out. In the path integral formulation it takes the form

$$\Delta_\sigma(\mu^2) = \frac{1}{\pi} \text{Im} \left[\Pi_{\text{shell}}^{(2)}(-s + i\epsilon) \right]_{s=\mu^2}, \quad (\text{E.1})$$

where $\Pi_{\text{shell}}^{(2)}$ is the self-energy projected onto the spin-2 channel with all internal lines restricted to the shell.

The classification below follows a strategy used in other many-body settings. In the work of Deng and collaborators on Hilbert's sixth problem [48], macroscopic fluid equations are derived from microscopic particle dynamics by classifying collision diagrams. Diagrams unprotected by conservation laws are suppressed in the thermodynamic limit. The protected class is resummed to yield the macroscopic equations. The same structural principle is used here. The microscopic objects are Feynman diagrams connecting fast and slow modes. The suppression mechanism is coarse-graining. The protected class is distinguished by momentum transfer.

All diagrams considered in this appendix contain only matter fields. Gravitational self-coupling does not appear because it arises only after the pole forms through Weinberg's low-energy theorem. See § 5.2 of the main text.

E.2 Feynman Diagrams Contributing to Δ_σ

The diagrams that contribute to Δ_σ are connected and irreducible. Their internal lines carry momenta in the shell $[1/\sigma, 1/(\sigma + d\sigma)]$. Their external legs carry slow-mode momenta below the shell. Energy-momentum tensor insertions occur at the vertices coupling to the external legs. These diagrams are classified by the momentum q transferred from the fast internal lines to the slow external legs.

E.3 Single Bubble Diagram

At one-loop order the diagram takes the form of a single bubble with two energy-momentum tensor insertions.

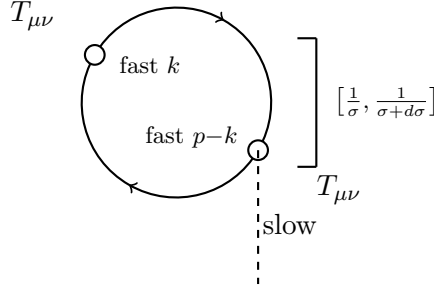


Figure 1. Single bubble diagram. The fermion loop carries momentum k (top arc, left to right) and $p - k$ (bottom arc, right to left), both restricted to the coarse-graining shell. Two energy-momentum tensor insertions $T_{\mu\nu}$ (filled circles) couple the fast loop modes to a slow external mode (dashed line). The infrared spectral behaviour is $\rho_0^{(2)}(s) \propto s^2$, contributing only to the continuum. No pole is produced at one-loop order.

This diagram yields

$$\rho_0^{(2)}(s) \propto s^2 \quad (s \rightarrow 0). \quad (\text{E.2})$$

It contributes only to the continuum spectrum and does not produce a pole. The corresponding single-bubble self-energy is denoted by $\Pi_0(s)$, with the infrared behaviour $\Pi_0(s) \propto s^2$.

E.4 Nonzero Momentum Transfer Diagrams

Diagrams with $q \neq 0$ involve a net momentum transfer from the fast modes to the slow external legs.

Under successive coarse-graining steps, each shell integration introduces an oscillatory factor $e^{iq \cdot x}$. In the infrared limit these oscillations average to zero. These diagrams contribute to the local noise kernel $\eta(t, \mathbf{x})$ in the Langevin equation. They do not contribute to the spectral pole.

E.5 Zero Momentum Transfer Ladder Diagrams

Diagrams with $q = 0$ take the form of fast modes entering in a back-to-back configuration with momenta k and $-k$.

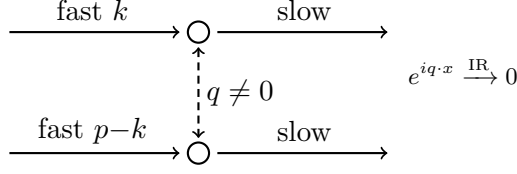


Figure 2. Nonzero momentum transfer diagram ($q \neq 0$). Each vertex converts a fast internal propagator into a slow external propagator, transferring net momentum $q \neq 0$. Under successive coarse-graining steps, the oscillatory factor $e^{iq \cdot x}$ averages to zero in the infrared limit. These diagrams contribute to the local noise kernel $\eta(t, \mathbf{x})$ and do not affect the spectral pole.

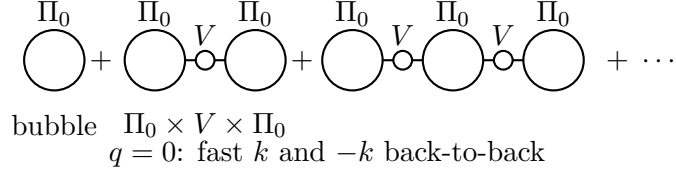


Figure 3. Zero momentum transfer ladder diagrams ($q = 0$). The geometric series resummation of single-bubble contributions Π_0 connected by irreducible insertions V . Each bubble contains fast modes in a back-to-back configuration with momenta k and $-k$, producing zero net momentum transfer. When the effective coupling λ_{eff} reaches $1/\Pi_0(0)$, the denominator of the geometric series vanishes and an isolated δ pole appears in the spectral function.

These diagrams do not oscillate. Their contributions can accumulate under coarse-graining. The resummation is a geometric series

$$\Delta_{\sigma}^{\text{ladder}} = \text{bubble} + \text{bubble} \times V \times \text{bubble} + \text{bubble} \times V \times \text{bubble} \times V \times \text{bubble} + \dots, \quad (\text{E.3})$$

where V denotes the irreducible insertions connecting the bubbles. The energy-momentum tensor is a conserved Noether current. Its matrix elements at $q = 0$ are not suppressed by any small parameter. The single-bubble contribution $\Pi_0(0)$ is of order $O(1)$. When the effective coupling λ_{eff} reaches the critical value $1/\Pi_0(0)$, the denominator of the geometric series vanishes. The spectral function acquires an isolated δ pole,

$$\rho^{(2)}(s) = Z\delta(s) + \rho_{\text{cont}}(s), \quad Z > 0. \quad (\text{E.4})$$

This gives a diagrammatic mechanism for the nonzero injection coefficient used in §§ 4.2 and 4.3 of the main text. The zero momentum transfer ladder class provides a cumulative contribution that is not washed out by coarse-graining. Within this class, a smooth spectral function cannot remain stationary once the cumulative threshold is reached.

E.6 Critical Threshold

Within the ladder resummation, a pole forms precisely when the geometric series reaches its critical threshold

$$1 - \lambda_{\text{eff}} \Pi_0(s) = 0 \quad \text{at } s = 0, \quad (\text{E.5})$$

where $\Pi_0(s)$ is the single-bubble contribution and λ_{eff} is the effective coupling of the irreducible insertions. This condition is model-dependent and non-perturbative. The non-vanishing of the energy-momentum tensor matrix elements at $q = 0$ follows from the conservation law and does not rely on perturbation theory. Whether λ_{eff} reaches the critical value at some coarse-graining scale depends on the full dynamical resummation of all relevant diagram classes in the specific model. This includes coupling strengths, particle spectra, and symmetry constraints. This is precisely the question that the spectral criterion is designed to answer.

The two classification criteria follow from energy-momentum conservation and diagram topology. A diagram with zero momentum transfer between fast and slow modes does not introduce an oscillatory factor under coarse-graining. Its contribution can accumulate. A diagram with nonzero momentum transfer introduces an oscillatory factor that averages to zero in the infrared limit. Among the diagrams with zero momentum transfer, the ladder diagrams formed by irreducible insertions connecting single bubbles are the class that can produce an isolated pole through geometric series resummation. The single bubble alone contributes only to the continuum. The conservation law guarantees that the matrix elements of the zero momentum transfer ladder diagrams are nonzero. This is the microscopic origin of the non-vanishing injection term. Whether the critical coupling is reached depends on the full dynamics of the specific model. This is the question that the spectral criterion addresses.

The classification by momentum transfer uses only the kinematic structure of the diagrams and the algebraic properties of conserved Noether currents. It does not depend on the order of perturbation theory or on the strength of the coupling. The same approach can be applied to other problems in which long-range correlations emerge from coarse-graining. The ladder diagram approximation is a standard tool for treating bound state formation in quantum field theory. Its reliability can be checked in solvable models such as the large- N model of Appendix B. In the Standard Model, non-ladder diagrams may contribute additional cumulative effects, modify the convergence condition of the geometric series, or shift the quantitative location of the critical threshold. They do not remove the kinematic fact that zero-momentum transfer diagrams supply a non-oscillatory injection channel. All diagrams considered in this appendix contain only matter fields. Gravitational self-coupling enters only after the pole forms through Weinberg’s low-energy theorem, as discussed in Section 5.2.

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Conflict of interest. The author declares no competing interests.

Data availability. Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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